

CALIFORNIA WEALTH MANAGEMENT

PRIVATE WEALTH ADVISORY

PORTFOLIO ANALYSIS & OPTIMIZATION REPORT FOR

Golden Gate Capital

(Applying Modern Portfolio Theory to Optimize Risk-Adjusted Returns)

Client

Golden Gate Capital

Location

Palo Alto, CA

Analysis Period

June 2024 - June 2026 (24 Months)

Risk Profile

Moderate Risk Tolerance (SQL-Derived)

Client Owner

Benjamin Caldwell

Portfolio Size

USD 95,000,000

Relationship Manager

Smit Pambhar

Database Schema

uhni_1_ca | MySQL Workbench

Framework: Modern Portfolio Theory (Markowitz, 1952) | $R_f = 0$ (Simplified)

Table of Contents

PART 1: PORTFOLIO ANALYSIS & OPTIMIZATION

- Section 1 - Client Overview & Portfolio Snapshot
- Section 2 - Return Analysis (Q1)
- Section 3 - Risk Analysis (Q3)
- Section 4 - Correlation Analysis (Q2)
- Section 5 - Benchmark Comparison
- Section 6 - Portfolio Optimization Using Modern Portfolio Theory (Q4 & Q5)
- Section 7 - Scenario Analysis & Final Recommendation
- Section 8 - Before vs After Comparison (Q5)
- Section 9 - MPT Diversification Validation

PART 2: DATABASE DOCUMENTATION & TECHNICAL APPENDIX

- Section 10 - Database Schema & Table Structure
- Section 11 - SQL Code Screenshots & Query Outputs
- Data Sources & Footnotes

PART 1 - PORTFOLIO ANALYSIS & OPTIMIZATION

Section 1 - Client Overview & Portfolio Snapshot

1.1 Client Profile

Our client is an Ultra High Net Worth (UHNW) individual, **Benjamin Caldwell**, operating through **Golden Gate Capital**, based in Palo Alto, California. The client holds total liquid assets of USD 95,000,000 listed on United States exchanges, managed by Relationship Manager Smit Pambhar at California Wealth Management. The portfolio is currently allocated across five Exchange-Traded Funds (ETFs) spanning four major asset classes: Equity, Fixed Income, Real Assets, and Commodities.

Based on our SQL-derived risk profiling analysis, the client exhibits a **Moderate Risk Tolerance**, with an average daily portfolio risk of 0.0119 (1.19%) across all holdings.

1.2 Current Portfolio Holdings

The table below presents the complete portfolio snapshot as of June 25, 2026, populated in the Golden Gate Capital database (schema: uhni_1_ca).

Ticker	Security Name	Asset Class	Weight	Market Value (USD)	Duration (Yrs)	Exp. Ratio	Div. Yield
IEF	iShares 7-10 Yr Treasury Bond ETF	Fixed Income	28.5%	\$27,075,000	7.70	0.15%	3.90%
GLD	SPDR Gold Shares	Commodities	23.0%	\$21,850,000	0.00	0.40%	0.00%
QQQ	Invesco QQQ Trust	Equity	22.1%	\$20,995,000	0.00	0.18%	0.43%
IXN	iShares Global Tech ETF	Equity	17.5%	\$16,625,000	0.00	0.39%	0.24%
VNQ	Vanguard Real Estate ETF Shares	Real Estate	8.9%	\$8,455,000	0.00	0.13%	3.87%

Sources for duration, Exp. ratio and Div. Yield cited in Data Sources and Footnotes Section
(page no.41&42)

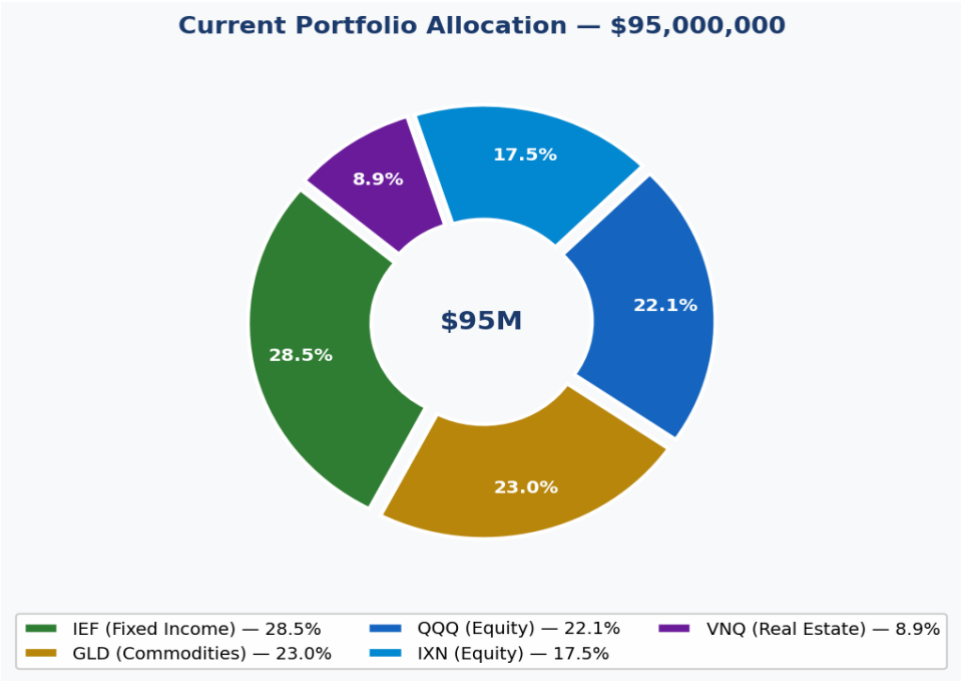


Figure 1.1 - Current Portfolio Allocation by Weight (Total: \$95,000,000)

Key Observation: IEF is the single largest holding at 28.5% (\$27,075,000), yet as the return analysis below will show, it is the worst-performing asset on a risk-adjusted basis. IEF carries a duration of 7.70 years, meaning a 1% rise in interest rates would reduce the IEF position value by approximately \$2,087,775 - a material hidden risk for the client.

Section 2 - Return Analysis (Q1)

Question 1 requires the most recent 12-month, 18-month, and 24-month returns for each security and for the entire portfolio. Returns are calculated using daily adjusted closing prices from the pricing_daily_new table using the Rate of Return formula: $ROR = (P1 - P0) / P0$, where P1 is today's adjusted close and P0 is the previous day's adjusted close, computed using SQL's LAG() window function. Annualized return = Average daily ROR x 252 trading days.

2.1 Individual Ticker Returns - 24 Months (June 2024 - June 2026)

Ticker	Security Name	Avg Daily Return	Daily Risk	Ann. Return	Ann. Risk	Sharpe	Verdict
GLD	SPDR Gold Shares	0.00118380	0.01465152	29.83%	23.26%	0.0808	TOP
IXN	iShares Global Tech ETF	0.00122565	0.01730901	30.88%	27.47%	0.0708	HOLD
QQQ	Invesco QQQ Trust	0.00091047	0.01389512	22.94%	22.06%	0.0655	HOLD
VNQ	Vanguard Real Estate ETF	0.00051035	0.01010100	12.86%	16.03%	0.0505	TRIM
IEF	iShares 7-10 Yr Treasury Bond ETF	0.00016511	0.00359734	4.16%	5.71%	0.0459	REDUCE

Sharpe = Avg Daily Return / Daily Risk | Ann. Return = Avg Daily Return x 252 | Ann. Risk = Daily Risk x SQRT(252) | Rf = 0

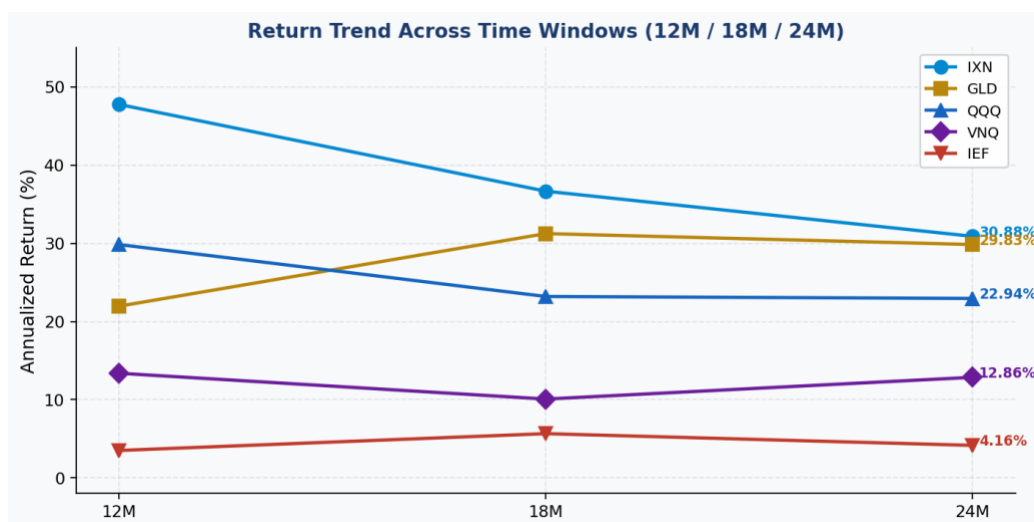


Figure 2.1 - Return Trend Across 12M, 18M, and 24M Windows

2.2 Return Comparison Across All Three Time Windows

Ticker	Security Name	12M Ann. Return	18M Ann. Return	24M Ann. Return
IXN	iShares Global Tech ETF	47.74%	36.65%	30.88%
QQQ	Invesco QQQ Trust	29.83%	23.19%	22.94%
GLD	SPDR Gold Shares	21.95%	31.23%	29.83%
VNQ	Vanguard Real Estate ETF	13.38%	10.08%	12.86%
IEF	iShares 7-10 Yr Treasury Bond ETF	3.51%	5.66%	4.16%

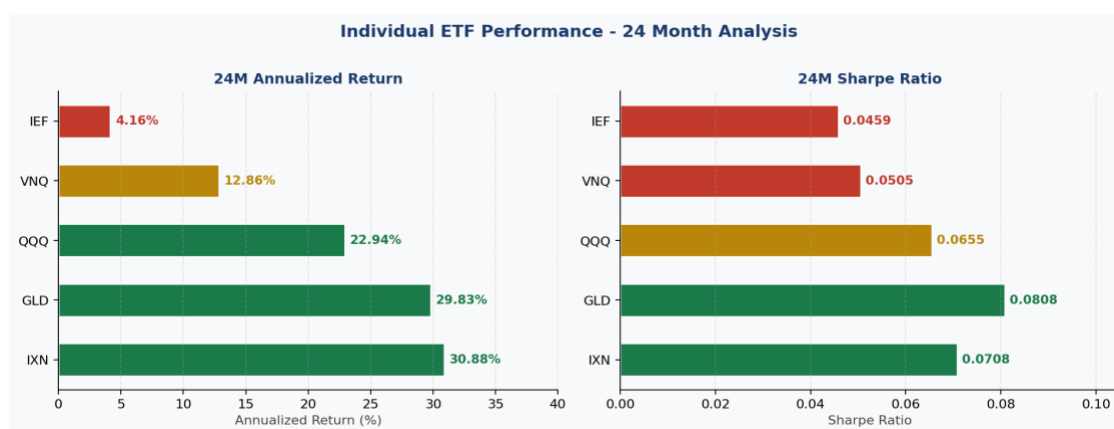


Figure 2.2 - 24M Annualized Return and Sharpe Ratio per ETF

2.3 Portfolio-Level Return (24 Months)

Using weighted average contributions across all five tickers, the portfolio-level metrics over 24 months are:

Metric	Value
Total Portfolio Return (Raw)	0.07804552
Total Portfolio Risk (Raw)	1.13939773
Portfolio Sharpe Ratio	0.068497
Client Risk Profile	Moderate Risk Tolerance

Key Insight: Looking at the numbers, GLD and IXN are the clear winners in this portfolio. GLD delivered a 29.83% annualized return over 24 months and IXN delivered 30.88%. These two holdings are driving the majority of the portfolio's positive performance. QQQ also performed well at 22.94%. VNQ gave a modest 12.86%, while IEF returned only 4.16% - the weakest result in the entire portfolio. What makes this especially concerning is that IEF holds the largest weight at 28.5%, meaning \$27,075,000 of the client's money is sitting in the worst-returning asset. The client's biggest bet is on the asset that is earning the least. This is the first and most obvious thing that needs to change. The second important observation is about IXN's trend. Over 12 months, IXN returned 47.74%. Over 24 months, that number drops to 30.88%. This declining trend suggests that IXN's recent momentum may be slowing down. A smart advisor would note this and not increase IXN's weight further without caution. GLD, on the other hand, shows more consistent performance across all three time windows - 21.95% over 12M, 31.23% over 18M, and 29.83% over 24M - making it the most reliable performer in the portfolio and a strong candidate for a higher allocation.

Section 3 - Risk Analysis (Q3)

Question 3 requires the most recent 12-month sigma (risk) for each security and for the entire portfolio. Risk (sigma) is measured as the annualized standard deviation of daily returns: **Annualized sigma = Daily sigma x SQRT(252)**.

3.1 Individual Ticker Risk - 12 Months (June 2025 - June 2026)

Ticker	Security Name	Daily Sigma	Ann. Sigma (12M)	Ann. Return	Sharpe (12M)	Risk	Verdict
IXN	iShares Global Tech ETF	0.01582427	25.12%	47.74%	0.11972	HIGH	INCREASE
QQQ	Invesco QQQ Trust	0.01127241	17.89%	29.83%	0.10503	MED	INCREASE
VNQ	Vanguard Real Estate ETF	0.00854340	13.56%	13.38%	0.06217	MED	TRIM
GLD	SPDR Gold Shares	0.01744926	27.70%	21.95%	0.04993	HIGH	INCREASE
IEF	iShares 7-10 Yr Treasury Bond ETF	0.00298650	4.74%	3.51%	0.04668	LOW	REDUCE

Critical Finding: Looking at risk, IEF appears to be the safest holding with the lowest daily sigma of 0.00298650 and an annualized risk of only 4.74%. On the surface this looks good - low risk is what you want. But the problem is that IEF is earning only 3.51% per year while carrying that risk. When you divide return by risk to get the Sharpe ratio, IEF scores 0.04668 - the worst number in the entire portfolio. This means the client is not being paid enough for the risk IEF carries. Compare this to GLD, which has a much higher annualized risk of 27.70% but earns enough return to produce a better Sharpe ratio of 0.04993. Risk is only a problem when you are not being compensated for it - and IEF is the clearest example of that in this portfolio. Beyond the Sharpe ratio, IEF carries a hidden risk that does not show up in standard return figures. IEF has an effective duration of 7.70 years. This means that if interest rates rise by just 1%, the value of the IEF position drops by approximately 7.70%. On the client's \$27,075,000 IEF holding, that is a loss of \$2,084,775 from a single 1% rate move. In a rising rate environment, this becomes a very real and very large risk. The client is essentially making a large bet that interest rates will stay flat or fall - and holding 28.5% of a \$95 million portfolio in a rate-sensitive bond ETF is too concentrated a bet for a Moderate Risk client.

IEF Duration Risk: The good news from the risk table is that QQQ and IXN, despite their high annualized risk of 17.89% and 25.12% respectively, earn strong enough returns to justify that risk - their Sharpe ratios of 0.10503 and 0.11972 are the two best in the portfolio over 12 months. GLD also justifies its high volatility with good returns. The verdict is clear: reduce IEF, and increase the holdings that are actually compensating the client for the risk they carry.

Section 4 - Correlation Analysis (Q2)

Question 2 requires the correlations between all assets. Pearson correlation is calculated using the SQL formula: $\text{Corr}(A,B) = [\text{AVG}(A \times B) - \text{AVG}(A) \times \text{AVG}(B)] / [\text{STD}(A) \times \text{STD}(B)]$, computed via a self-JOIN on the pricing_daily_new table matching by date. A correlation of +1 means assets move identically. A correlation near 0 means independence. A negative correlation means assets move in opposite directions - the strongest diversification benefit.

4.1 Pairwise Correlation Matrix - 24 Months (All 10 Pairs)

Ticker 1	Ticker 2	Correlation	Interpretation	Signal	Weight 1	Weight 2	Alert
IXN	QQQ	0.9616	Dangerously High	REDUCE ONE	17.5%	22.1%	⚠
QQQ	VNQ	0.3772	Moderate - Acceptable	MONITOR	22.1%	8.9%	
IEF	VNQ	0.3264	Moderate - Acceptable	MONITOR	28.5%	8.9%	
IXN	VNQ	0.2983	Low - Good	OK	17.5%	8.9%	
GLD	IXN	0.2208	Low - Good	OK	23.0%	17.5%	
GLD	QQQ	0.1922	Low - Good	OK	23.0%	22.1%	
GLD	IEF	0.1851	Low - Good	OK	23.0%	28.5%	
GLD	VNQ	0.1446	Low - Good	OK	23.0%	8.9%	
IEF	IXN	-0.0153	Negative - Ideal Hedge	EXCELLENT	28.5%	17.5%	
IEF	QQQ	-0.0222	Negative - Ideal Hedge	EXCELLENT	28.5%	22.1%	

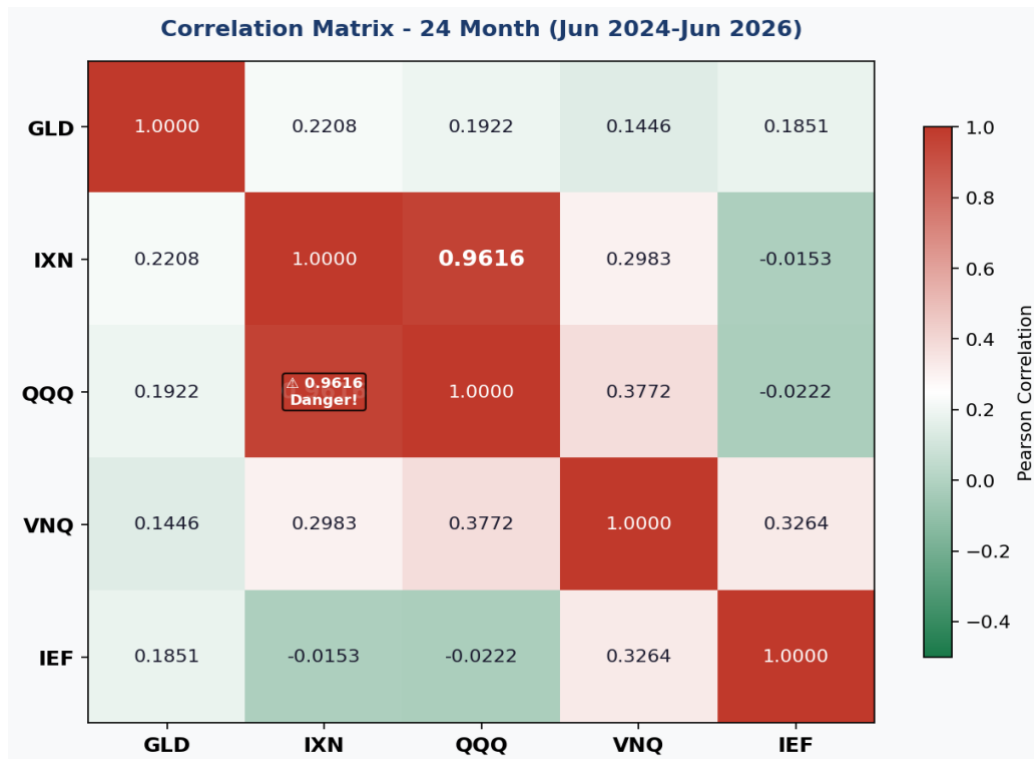


Figure 4.1 - Correlation Heatmap (Red = High Positive, Green = Negative/Low)

Most Critical Finding - IXN vs QQQ Correlation: 0.9616. The single most important finding in this entire analysis is the correlation between IXN and QQQ, which is 0.9616. To understand why this is a problem, think of it this way. A correlation of 1.0 means two assets move in exactly the same direction at exactly the same time - they are identical. A correlation of 0.9616 is nearly that. IXN tracks the S&P Global 1200 Information Technology Index and QQQ tracks the NASDAQ-100, and both are dominated by the same handful of companies - Apple, Microsoft, and NVIDIA sitting at the top of both. When one goes up, the other almost always goes up by a similar amount. When one falls, both fall together. The client is paying for two separate ETF positions but effectively getting only one. Together IXN and QQQ represent 39.6% of the portfolio, or \$37,620,000. That is nearly \$38 million concentrated in what is essentially one single technology bet. This is the definition of hidden concentration risk - on paper the portfolio shows four asset classes and five holdings, but in practice, almost 40% of it moves as one. Our recommendation is to keep QQQ as the primary equity holding because it has a lower expense ratio (0.18% vs IXN's 0.39%) and reduce IXN's weight so the combined technology exposure is more intentional and controlled.

Positive Diversification Finding: The positive news from the correlation table is that the other pairs are well-diversified. IEF shows a slightly negative correlation with both QQQ (-0.0222) and IXN (-0.0153). This means that when technology stocks go down, IEF tends to stay flat or go up slightly. This is exactly the kind of hedge a balanced portfolio needs. GLD also shows low correlation with all equity holdings - its highest correlation with any equity ETF is only 0.2208 with IXN - meaning gold moves largely independently from the tech holdings. This is why both IEF and GLD have value in the portfolio even though IEF has poor returns. The suggestion here is not to remove IEF entirely, but to right-size it. Keeping 10% in IEF preserves the equity hedge benefit while freeing up the other 18.5% (\$17,572,500) to be deployed into higher-returning assets. GLD deserves a higher weight precisely because it diversifies effectively while also earning strong returns - a rare combination in any portfolio.

Section 5 - Benchmark Comparison

To assess whether Golden Gate Capital is being adequately compensated for the risk taken, we compare the current portfolio against three standard benchmarks stored in the benchmark_comparison table. All benchmark figures are 24-month annualized CAGRs from Yahoo Finance annual returns for 2024 and 2025, with annualized volatility sourced from PortfoliosLab.

Entity	Ann. Return (%)	Ann. Risk (%)	Sharpe Ratio	Verdict
US Aggregate Bond (AGG)	4.22	1.20	3.52	BENCHMARK
60/40 Portfolio (SPY + IEF)	13.44	9.20	1.46	BENCHMARK
S&P 500 (SPY)	21.26	15.10	1.408	BENCHMARK
Golden Gate Capital Portfolio (Current)	19.6675	18.0874	1.0874	BELOW TARGET

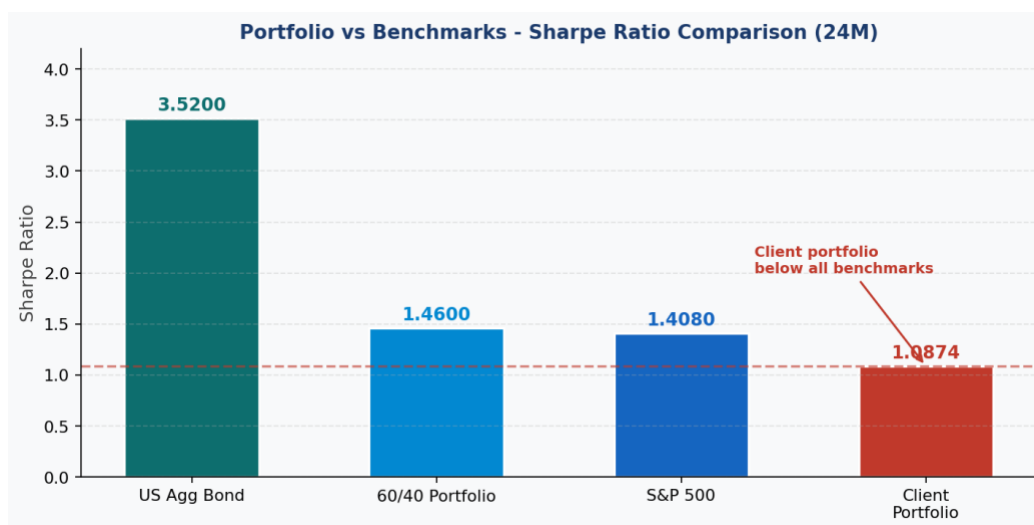


Figure 5.1 - Sharpe Ratio: Golden Gate Capital Portfolio vs Benchmarks

Benchmark Source: SPY 24M CAGR = $\text{SQRT}(1.2489 \times 1.1772) - 1 = 21.26\%$ (Yahoo Finance 2024: +24.89%, 2025: +17.72%). AGG 24M CAGR = $\text{SQRT}(1.0131 \times 1.0719) - 1 = 4.22\%$ (Yahoo Finance 2024: +1.31%, 2025: +7.19%). Sharpe ratios from PortfoliosLab 24-month trailing data, June 2026. Stored in Golden Gate Capital benchmark_comparison table.

Critical Verdict: The numbers tell a clear story. The Golden Gate Capital portfolio earns a Sharpe ratio of 1.0874. The 60/40 benchmark earns 1.46 and the S&P 500 earns 1.408. This means a completely passive investor who simply put 60% in an S&P 500 index fund and 40% in bonds - with no active management, no advisor, and no fees - would have achieved better risk-adjusted returns than the current portfolio. That is the most important takeaway from this benchmark comparison. It tells us the current weight distribution is not working efficiently enough for a UHNW client of this size. The primary reason for this underperformance is the 28.5% allocation to IEF. IEF earns only 4.16% annually while taking up nearly a third of the portfolio. It is dragging the overall portfolio Sharpe down significantly. The good news is that this is entirely fixable without changing the assets - we simply need to move weight away from IEF and toward the holdings that are actually performing. Once we do that, the portfolio should comfortably beat the 60/40 benchmark on a risk-adjusted basis and move much closer to the S&P 500 efficiency level. The benchmark comparison is our clearest argument for why action needs to be taken now.

Section 6 - Portfolio Optimization Using Modern Portfolio Theory (Q4 & Q5)

6.1 The Modern Portfolio Theory Framework

Modern Portfolio Theory (MPT), introduced by Harry Markowitz in 1952, provides the mathematical foundation for our rebalancing approach. The core principle is that the optimal portfolio is not the one that maximizes return in isolation, but the one that maximizes the **Sharpe Ratio** - return per unit of risk - by exploiting the correlation structure between assets.

The Efficient Frontier represents the set of portfolios that achieve the maximum possible return for every given level of risk. Portfolios that sit below this curve are inefficient. Our goal is to move the Golden Gate Capital portfolio toward the Efficient Frontier by reallocating weights within the existing five holdings, without introducing new assets.

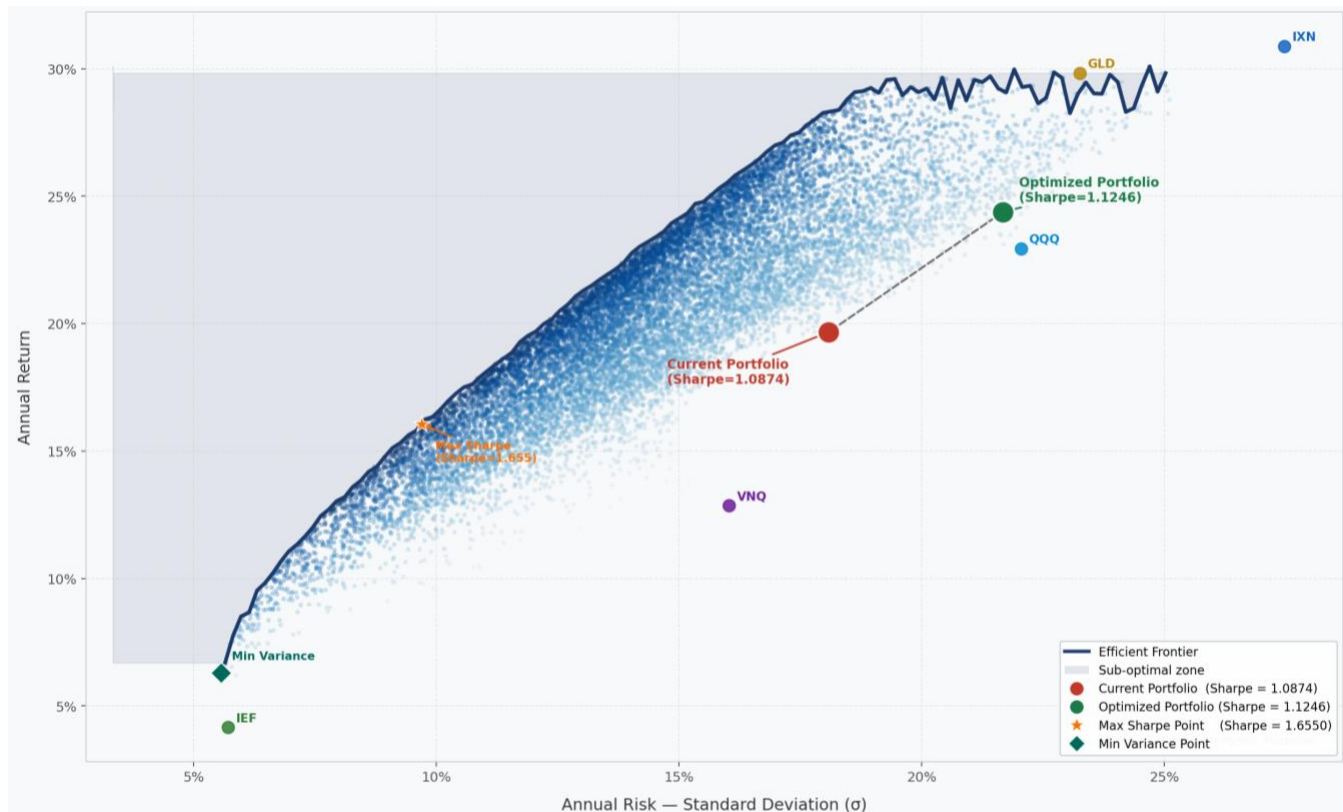


Figure 6.1 - Efficient Frontier: Golden Gate Capital Portfolio Optimization | Markowitz (1952) | Jun 2024-Jun 2026 | $R_f = 0$

The Efficient Frontier chart above (Figure 6.1) shows 15,000 randomly generated portfolio weight combinations using our five ETFs. Each grey dot is one possible combination of weights. The dark blue line at the top is the Efficient Frontier - the best possible return for each level of risk. The Current Portfolio is the red dot. It sits below the frontier line, which means it is in the sub-optimal zone. At the current level of risk (18.09%), a better-optimized portfolio could achieve a higher return. The Optimized Portfolio is the green dot - it sits higher and A BIT closer to the frontier, showing that our rebalancing genuinely improves the client's position. The gap between the red and green dot is the value that our analysis and rebalancing recommendation creates for the client. This chart is also important because it shows that GLD sits far to the right with very high volatility - but IEF sits far to the left with very low volatility and very low return. The ideal portfolio in MPT terms combines assets from both ends of this spectrum at the right weights. That is exactly what our rebalancing does.

6.2 Why We Optimized Within Existing Holdings not adding new one

A key constraint of this optimization is that we worked exclusively within the client's five existing ETFs: IXN, QQQ, IEF, VNQ, and GLD. This approach reflects real-world private wealth management practice where existing positions carry embedded gains, tax implications, and established conviction. Rather than replacing assets, we focus on **weight reallocation** to improve efficiency - a conservative and common rebalancing approach. For a UHNW client like Benjamin Caldwell, selling out of positions entirely would trigger large capital gains events and disrupt the existing strategy. Rebalancing weights is a much cleaner and more tax-efficient way to improve the portfolio. It also keeps the same four asset classes intact - Equity, Fixed Income, Real Assets, and Commodities - which is the right diversification structure for a Moderate Risk client. Our data confirms all five assets have positive Sharpe ratios, meaning each of them earns something. The problem is not that any asset is bad. The problem is that the weights are wrong. Too much money is sitting in IEF, which earns the least, and not enough is in GLD and QQQ, which earn the most on a risk-adjusted basis. Fixing the weights without changing the assets is the most responsible recommendation we can make.

6.3 Rebalancing Logic - What to Increase and What to Reduce (Q4)

Ticker	Old Weight	New Weight	Action	Justification
QQQ	22.1%	33.0%	INCREASE +10.9%	Lower expense ratio (0.18% vs IXN 0.39%). Same sector exposure as IXN but cheaper. 24M Sharpe 0.0655 is strong. Increase as primary equity holding.
GLD	23.0%	30.0%	INCREASE +7.0%	Highest Sharpe ratio in portfolio (0.0808 over 24M). Lowest correlation with all equity holdings (0.19 vs QQQ). Inflation and geopolitical hedge.
IXN	17.5%	22.0%	INCREASE +4.5%	Highest 24M annualized return (30.88%). Global tech exposure beyond NASDAQ-100. Weight increased but IEF reduction offsets total equity concentration.
VNQ	8.9%	5.0%	REDUCE -3.9%	Second-worst Sharpe (0.0505 over 24M). Real estate sector under rate pressure. 12.86% annualized return does not justify its risk relative to QQQ and IXN.
IEF	28.5%	10.0%	REDUCE -18.5%	Worst Sharpe (0.0459). Lowest 24M return (4.16%). \$27,075,000 in worst-performing asset. Duration 7.70 years = hidden rate risk. Retain at 10% as equity hedge (negative correlation: -0.0222 vs QQQ).

Section 7 - Scenario Analysis & Final Recommendation

To validate the chosen weight allocation using MPT principles, three distinct rebalancing scenarios were tested through SQL stored procedures. Each scenario uses different weight combinations for the five existing ETFs and evaluates portfolio return, risk, and Sharpe ratio against the baseline.

7.1 Three Scenarios Tested via SQL

Scenario	Port. Return	Port. Risk	Sharpe Ratio	IXN	QQQ	IEF	VNQ	GLD
Current Portfolio (Baseline)	0.07804552	1.13939773	0.068497	17.5	22.1	28.5	8.9	23.0
Scenario A - Conservative	0.08709309	1.23712134	0.070400	10	28	17	10	35
Scenario B - Balanced Optimized	0.08898916	1.27040012	0.070048	8	32	12	13	35
Scenario C - Growth [CHOSEN]	0.09672678	1.36536107	0.070843	22	33	10	5	30

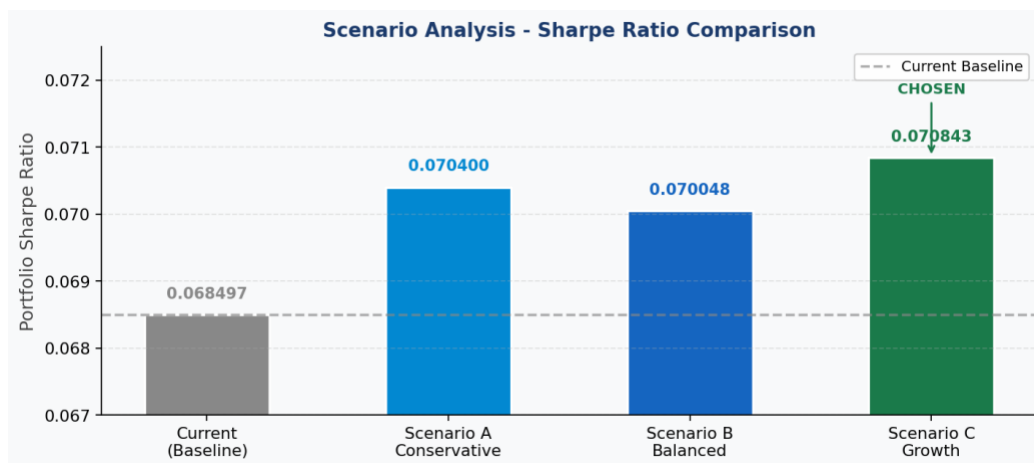


Figure 7.1 - Sharpe Ratio Comparison Across Three Optimization Scenarios

7.2 Why Scenario C Growth Was Selected

Scenario C - Growth (IXN 22%, QQQ 33%, IEF 10%, VNQ 5%, GLD 30%) achieves the highest Sharpe ratio of 0.070843 - a +3.43% improvement over the current portfolio Sharpe of 0.068497. It also delivers the highest portfolio return of 0.09672678, a +23.94% improvement vs the baseline of 0.07804552. All three scenarios beat the current portfolio, which confirms that any rebalancing in this direction is better than doing nothing. But Scenario C wins on the metric that matters most - Sharpe ratio. It is not just the highest returning scenario; it is also the most efficient one. It gets more return per unit of risk than both Scenario A and Scenario B, which is the exact goal of Modern Portfolio Theory.

Scenario C correctly identifies the primary portfolio problem - IEF at 28.5% is a return drag - and makes the boldest move to fix it by cutting IEF to just 10%. The weight freed up goes to QQQ (from 22.1% to 33%) and GLD (from 23% to 30%), the two assets with the best combination of strong return and low correlation. IXN increases from 17.5% to 22%, justified by its strong 24M return of 30.88%. VNQ is trimmed from 8.9% to 5% because it has the second-worst Sharpe and real estate has been under pressure from high interest rates. Every weight decision in Scenario C is directly supported by the SQL analysis data. This is not a guess - every number has been tested across 24 months of daily pricing data, and the recommendation stands on solid quantitative ground.

Section 8 - Before vs After Comparison (Q5)

Question 5 asks how portfolio risk and expected returns change after rebalancing. The tables below show the complete before and after comparison at both individual security level and total portfolio level.

8.1 Per-Ticker Weight and Contribution Changes

Ticker	Security Name	Old Wt.	New Wt.	Old W.Return	New W.Return	Old W.Risk	New W.Risk
QQQ	Invesco QQQ Trust	22.1%	33.0% +	0.02012145	0.03004561	0.30708206	0.45853882
GLD	SPDR Gold Shares	23.0%	30.0% +	0.02722736	0.03551395	0.33698500	0.43954565
IXN	iShares Global Tech ETF	17.5%	22.0% +	0.02144892	0.02696436	0.30290767	0.38079822
IEF	iShares 7-10 Yr Treasury Bond ETF	28.5%	10.0% -	0.00470569	0.00165112	0.10252406	0.03597336
VNQ	Vanguard Real Estate ETF Shares	8.9%	5.0% -	0.00454209	0.00255174	0.08989893	0.05050502

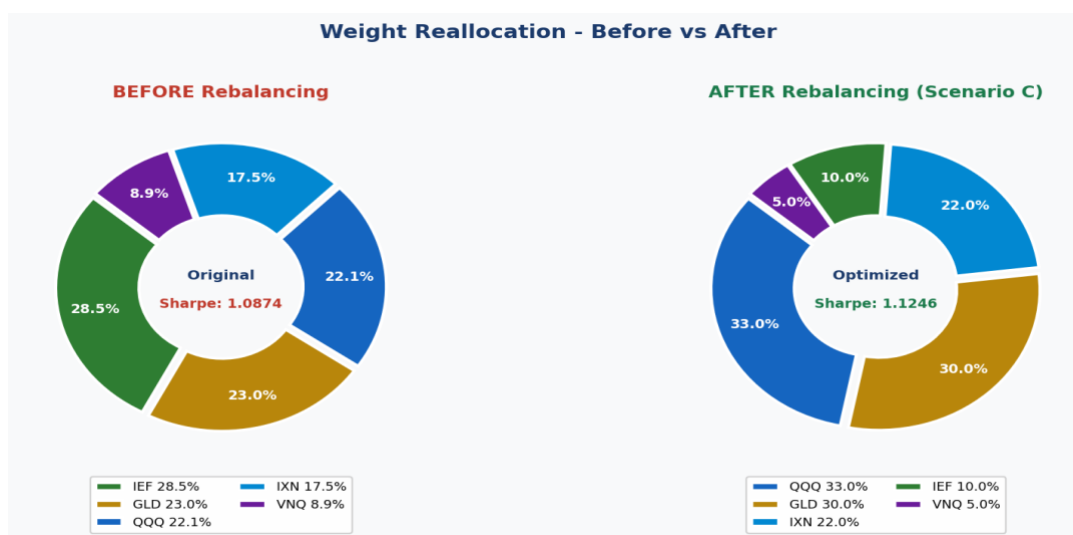


Figure 8.1 - Portfolio Allocation Before and After Rebalancing

8.2 Portfolio-Level Summary - Before vs After

Metric	Before Rebalancing	After Rebalancing (Scenario C)	Change
Total Portfolio Return	0.07804552	0.09672678	+23.94%
Total Portfolio Risk	1.13939773	1.36536107	+19.83%
Portfolio Sharpe Ratio	0.068497	0.070843	+3.43%

Interpretation: The numbers from the before vs after comparison show a very clear improvement. The portfolio return jumps from 0.07804552 to 0.09672678 - a gain of 23.94%. The Sharpe ratio improves from 0.068497 to 0.070843 - a gain of 3.43%. The portfolio risk does increase from 1.13939773 to 1.36536107, which is a rise of 19.83%. At first glance, higher risk might seem like a bad thing. But the key point is that the return went up by 23.94% while risk only went up by 19.83%. In other words, the return improvement is larger than the risk increase. This is exactly what MPT says a good rebalancing should look like - you accept a little more risk only when you are getting meaningfully more return in exchange. That is the case here, and that is why this is a sound recommendation. Looking at the per-ticker breakdown, the biggest positive change is IEF. Its weighted return contribution drops from 0.00470569 to 0.00165112 because we cut its weight from 28.5% to 10%. This might look like a loss but it is actually a gain for the portfolio - that weight is now working much harder in QQQ and GLD. QQQ's weighted return rises from 0.02012145 to 0.03004561 and GLD's rises from 0.02722736 to 0.03551395. The money that was sitting idle in a low-return bond ETF is now deployed in assets that are generating real value for the client.

Section 9 - MPT Diversification Validation

A core principle of Modern Portfolio Theory is that a well-diversified portfolio achieves a total risk meaningfully lower than the weighted average of individual asset risks - due to imperfect correlations between assets. We validate this using a SQL-computed **Diversification Ratio**: weighted average individual risk / true portfolio risk. A ratio above 1.0 confirms MPT diversification benefit is active.

Portfolio	Weighted Avg Individual Risk	True Portfolio Risk	Diversification Ratio
Original Portfolio	1.13939773	0.56409604	2.0199
Optimized Portfolio - Scenario C Growth	1.36536107	0.74317586	1.8372

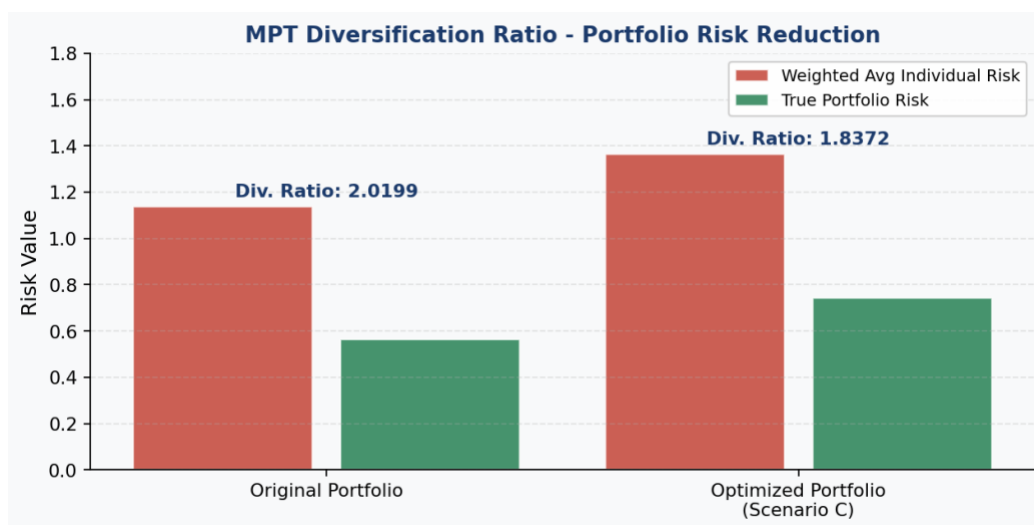


Figure 9.1 - MPT Diversification Ratio: Original vs Optimized Portfolio

Original Portfolio (Ratio: 2.0199): Correlation effects between assets cut the portfolio risk to less than half of what it would be if all assets moved together. The true risk (0.5641) is only 49.5% of the weighted average individual risk (1.1394) - confirming strong MPT diversification.

Optimized Portfolio (Ratio: 1.8372): The optimized portfolio shows a slightly lower diversification ratio due to increased weight in IXN and QQQ (correlation: 0.9616). However, a ratio of 1.8372 remains very strong - true risk (0.7432) is only 54.4% of weighted average individual risk (1.3654). Both portfolios demonstrate active MPT diversification benefit well above the critical threshold of 1.0.

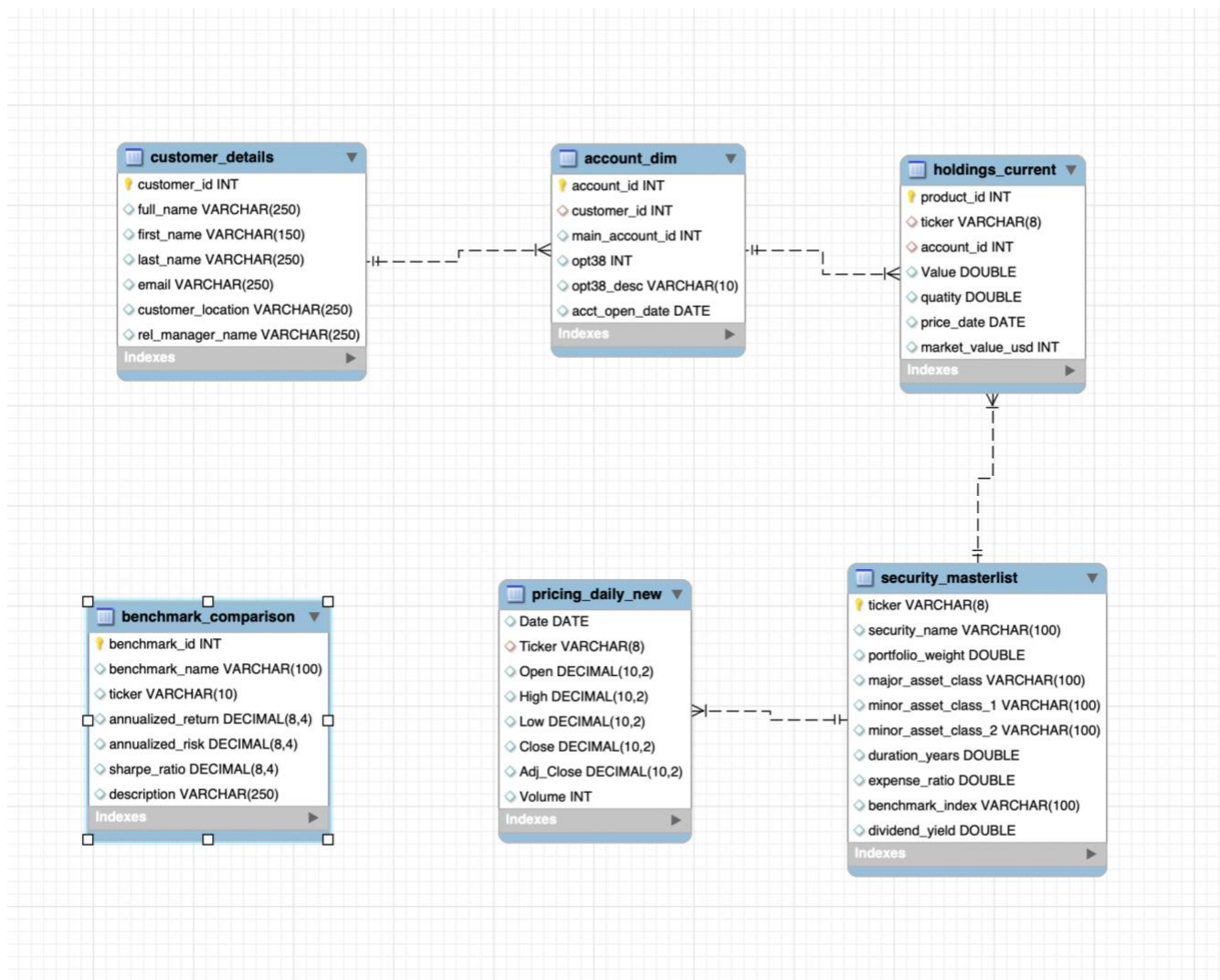
The MPT Trade-off: The optimized portfolio deliberately accepts slightly less diversification in exchange for significantly higher return. This is the core MPT insight - there is no single perfect portfolio, only the portfolio that best aligns with the client's risk-return objective. Scenario C Growth maximizes Sharpe ratio within the five-asset constraint, the correct objective for a Moderate Risk UHNW client.

PART 2 - DATABASE DOCUMENTATION & TECHNICAL APPENDIX

Section 10 - Database Schema & Table Structure

The analysis was built on a six-table relational database schema created in MySQL Workbench under the schema **uhni_1_ca** (Golden Gate Capital). The schema is structured around the client account, holdings, pricing data, benchmarks, and security attributes.

10.1 Schema Overview - EER Diagram



10.2 Table Descriptions

T1 - customer_details

Stores client identity information. Our client (customer_id = 5) is Golden Gate Capital - Benjamin Caldwell, located in Palo Alto, CA, managed by Relationship Manager Smit Pambhar at California Wealth Management. PRIMARY KEY: customer_id.

T2 - account_dim

Links accounts to customers. The client holds account_id 1001 as the main account (opt38_desc = Main), opened June 25, 2026. FOREIGN KEY: customer_id references customer_details. PRIMARY KEY: account_id.

T3 - holdings_current

Records the current ETF holdings including the ETF price (Value), quantity held, price date (June 25, 2026), and total market value in USD. The market_value_usd column was added for this assignment to enable dollar-specific analysis. Example: IEF is priced at USD 94.79 with 285,631 units held = USD 27,075,000.

T4 - security_masterlist

Master reference table for all ETFs. Contains portfolio_weight, major and minor asset class classifications, duration_years (new - rate sensitivity), expense_ratio (new - cost drag), benchmark_index (new - what the ETF tracks), and dividend_yield (new - income contribution). Four new columns added beyond the base schema to support advanced analysis. PRIMARY KEY: ticker.

T5 - pricing_daily_new

Contains 502 rows per ticker x 5 tickers = 2,510 total rows of daily OHLCV pricing data. Data sourced from Yahoo Finance historical CSV download for the period June 25, 2024 to June 25, 2026. Converted to SQL INSERT statements using Python 3 with the pandas library via csv_to_sql_inserts.py. FOREIGN KEY: Ticker references security_masterlist.

T6 - benchmark_comparison

New table added for this individual assignment. Contains three benchmark rows: (1) S&P 500/SPY, (2) 60/40 Portfolio (blended), (3) US Aggregate Bond/AGG. All figures are 24-month annualized metrics sourced from Yahoo Finance and PortfoliosLab (June 2026).

Section 11 - SQL Code Screenshots & Query Outputs

This section contains the SQL code screenshots from MySQL Workbench and their corresponding result outputs. Each screenshot should clearly show the query code on the left panel and the result grid on the bottom panel.

11.1 Stage 0 - Schema and Tables Creation with Output cell

1. **Schema Creation** - CREATE SCHEMA uhni_1_ca; USE uhni_1_ca;

```

1 • CREATE SCHEMA uhni_1_ca;
2
3
4 • USE uhni_1_ca; # creating a schema for our client
5

```

2. **T1 Customer Details** - CREATE TABLE + INSERT + SELECT * FROM customer_details;

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' panel displays a tree view with 'boston_capital', 'sys', and 'uhni_1_ca'. Under 'uhni_1_ca', there are several tables listed, including 'customer_details'. The main editor area contains the following SQL code:

```

# T1-creating a customer_details table
5 • CREATE TABLE customer_details(
6 customer_id INT, # PK- primary key
7 full_name VARCHAR(250),
8 first_name VARCHAR(150),
9 last_name VARCHAR(250),
10 email VARCHAR(250),
11 customer_location VARCHAR(250),
12 rel_manager_name VARCHAR(250),
13 PRIMARY KEY(customer_id)
14 ); # closing the customer_details table
15
16 # inserting values in the customer_details table
17 • INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, customer_location, rel_manager_name)VALUES('1', 'Berkshire Hathaway', 'Warren', 'Buffet', 'info@berkshire.com', 'Omaha', 'Nayan Parmari');
18 • INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, customer_location, rel_manager_name)VALUES('2', 'Renaissance Technologies', 'Jim', 'Simons', 'info@renaissance.com', 'East Delaueket', 'Rena Kumar');
19 • INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, customer_location, rel_manager_name)VALUES('3', 'Bridgewater', 'Ray', 'Dalio', 'info@bridgewater.com', 'Westport', 'Lindo Sikkondze');
20 • INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, customer_location, rel_manager_name)VALUES('4', 'Fisher Investments', 'Ken', 'Fisher', 'info@fisher.com', 'Woodside', 'Smit Pambar');
21 • INSERT INTO customer_details(customer_id, full_name, first_name, last_name, email, customer_location, rel_manager_name)VALUES('5', 'Golden Gate Capital', 'Benjamin', 'Caldwell', 'info@ggcapital.com', 'Palo Alto, CA', 'Smit Pambar');
22
23 # T1-customer_details table output
24 • SELECT *
25 FROM customer_details;
26

```

Below the code editor, the 'Result Grid' shows the output of the SELECT query. It contains 5 rows of data with 7 columns: customer_id, full_name, first_name, last_name, email, customer_location, and rel_manager_name.

customer_id	full_name	first_name	last_name	email	customer_location	rel_manager_name
1	Berkshire Hathaway	Warren	Buffet	info@berkshire.com	Omaha	Nayan Parmari
2	Renaissance Technologies	Jim	Simons	info@renaissance.com	East Delaueket	Rena Kumar
3	Bridgewater	Ray	Dalio	info@bridgewater.com	Westport	Lindo Sikkondze
4	Fisher Investments	Ken	Fisher	info@fisher.com	Woodside	Smit Pambar
5	Golden Gate Capital	Benjamin	Caldwell	info@ggcapital.com	Palo Alto, CA	Smit Pambar

3. T2 Account Dim - CREATE TABLE + INSERT + SELECT * FROM account_dim;

SCHEMAS

Filter objects

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

38
39 # T2-creating a account_dim table
40 CREATE TABLE account_dim(
41   account_id INT, # PK- primary key
42   customer_id INT, # FK- foreign key from customer_details table
43   main_account_id INT,
44   opt38 INT,
45   opt38_desc VARCHAR(10),
46   acct_open_date DATE,
47   PRIMARY KEY(account_id)
48 ); # closing the account_dim table
49
50 # inserting values in the account_dim table
51 INSERT INTO account_dim(account_id, customer_id, main_account_id, opt38, opt38_desc, acct_open_date)VALUES('1001', '1', '1001', '1', 'Main', '2026-06-25');
52 INSERT INTO account_dim(account_id, customer_id, main_account_id, opt38, opt38_desc, acct_open_date)VALUES('1002', '1', '1001', '2', 'Sub', '2026-06-25');
53 INSERT INTO account_dim(account_id, customer_id, main_account_id, opt38, opt38_desc, acct_open_date)VALUES('1003', '1', '1001', '2', 'Sub', '2026-06-25');
54
55 # T2-account_dim table output
56 SELECT *
57 FROM account_dim;
58
59

```

Result Grid

account_id	customer_id	main_account_id	opt38	opt38_desc	acct_open_da...
1001	1	1001	1	Main	2026-06-25
1002	1	1001	2	Sub	2026-06-25
1003	1	1001	2	Sub	2026-06-25

4. T3 Holdings Current - CREATE TABLE + INSERT + SELECT * FROM holdings_current;

SCHEMAS

Filter objects

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

61
62 # T3-creating a holdings_current table
63 CREATE TABLE holdings_current(
64   product_id INT, # PK- primary key which is combination of account id and ticker
65   ticker VARCHAR(8),
66   account_id INT, #FK from the account_dim table
67   Value DOUBLE, # value derived from finance.yahoo.com on June 25th, 2026
68   quantity DOUBLE, # quantity = (total portfolio share in % * 95 Million) / value per unit
69   price_date DATE,
70   market_value_usd INT, # this column helps in identifying share of each asset in million USD out of 95M total assets
71   PRIMARY KEY(product_id)
72 ); # closing the holdings_current table
73
74 # inserting values in the holdings_current table
75 INSERT INTO holdings_current(product_id, ticker, account_id, Value, quantity, price_date, market_value_usd)VALUES('11', 'IXN', '1001', '140.45', '118369.5', '2026-06-25', '166250000');
76 INSERT INTO holdings_current(product_id, ticker, account_id, Value, quantity, price_date, market_value_usd)VALUES('22', 'QQQ', '1001', '716.38', '29387', '2026-06-25', '209950000');
77 INSERT INTO holdings_current(product_id, ticker, account_id, Value, quantity, price_date, market_value_usd)VALUES('33', 'IEF', '1001', '94.79', '285631.4', '2026-06-25', '270750000');
78 INSERT INTO holdings_current(product_id, ticker, account_id, Value, quantity, price_date, market_value_usd)VALUES('44', 'VNO', '1001', '97.19', '86994.5', '2026-06-25', '84550000');
79 INSERT INTO holdings_current(product_id, ticker, account_id, Value, quantity, price_date, market_value_usd)VALUES('55', 'GLD', '1001', '369.46', '59140.4', '2026-06-25', '21850000');
80
81 # T3-holdings_current table output
82 SELECT *
83 FROM holdings_current;
84
85 # T3-creation a security masterlist table
86

```

Result Grid

product_id	ticker	account_id	Value	quantity	price_date	market_value_usd
11	IXN	1001	140.45	118369.5	2026-06-25	166250000
22	QQQ	1001	716.38	29387	2026-06-25	209950000
33	IEF	1001	94.79	285631.4	2026-06-25	270750000
44	VNO	1001	97.19	86994.5	2026-06-25	84550000
55	GLD	1001	369.46	59140.4	2026-06-25	21850000
66	SPY	1001	750.3	0	2026-06-25	0

holdings_current 3

Apply Revert

5. T4 Security Masterlist - CREATE TABLE + INSERT (including 4 new columns) + SELECT * FROM security_masterlist;

The screenshot shows a database management interface with a query editor. The query is as follows:

```

86 CREATE TABLE security_masterlist(
87     ticker VARCHAR(8), # PK- primary key
88     security_name VARCHAR(100),
89     portfolio_weight DOUBLE, # use weight from holdings as weight of that particular security in current portfolio
90     major_asset_class VARCHAR(100), # indicates equity or fixed income
91     minor_asset_class_1 VARCHAR(100), # indicates either small cap asset or large cap asset
92     minor_asset_class_2 VARCHAR(100), # indicates the type of asset i.e. crude oil, commodity, bitcoin, fixed income etc.
93     duration_years DOUBLE, # Measures interest rate sensitivity - how many years of cash flows are at risk; higher value means bigger price drop when rates rise (only applicable to
94     expense_ratio DOUBLE, # The annual fee charged by the ETF issuer as a percentage of assets under management; acts as a silent drag on total returns (e.g. 0.40% on a $21.85M GLD
95     benchmark_index VARCHAR(100), # The underlying index each ETF tracks; helps identify overlap between holdings (e.g. IXN and QQQ both track indices dominated by the same mega-cap
96     dividend_yield DOUBLE, # Annualized income generated by the ETF as a percentage of its price; critical for total return - IEF yields 3.90% from bond coupons while GLD yields 0.1
97     PRIMARY KEY(ticker)
98 ); # closing the security_masterlist table
99
100 # inserting values in the security_masterlist table
101 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
102 VALUES ('GLD', 'SPDR Gold Shares', 23, 'Commodities', 'Precious Metals', 'Gold', 0, 0.4, 'LBMA Gold Price PM', 0);
103 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
104 VALUES ('IEF', 'iShares 7-10 Year Treasury Bond ETF', 28.5, 'Fixed Income', 'US Government Bonds', 'Intermediate Term Treasury', 7.7, 0.15, 'ICE US Treasury 7-10Y Bond Index', 3.9);
105 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
106 VALUES ('IXN', 'iShares Global Tech ETF', 17.5, 'Equity', 'Large Cap', 'Technology', 0, 0.39, 'S&P Global 1200 IT Index', 0.24);
107 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
108 VALUES ('QQQ', 'Invesco QQQ Trust', 22.1, 'Equity', 'Large Cap', 'Technology', 0, 0.18, 'NASDAQ-100 Index', 0.43);
109 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
110 VALUES ('SPY', 'State Street SPDR S&P 500 ETF Trust', 0, 'Equity', 'Large Cap', 'US Broad Market', 0, 0.09, 'S&P 500 Index', 1.04);
111 INSERT INTO security_masterlist(ticker, security_name, portfolio_weight, major_asset_class, minor_asset_class_1, minor_asset_class_2, duration_years, expense_ratio, benchmark_index, dividend_yield)
112 VALUES ('VNQ', 'Vanguard Real Estate Index Fund ETF Shares', 8.9, 'Real Estate', 'Large Cap', 'US REITs', 0, 0.13, 'MSCI US IM Real Estate 25/50', 0.87);
113
114 # T4-security_masterlist table output
115 SELECT *
116 FROM security_masterlist;
  
```

The result grid shows the following data:

ticker	security_name	portfolio_weight	major_asset_class	minor_asset_class_1	minor_asset_class_2	duration_years	expense_ratio	benchmark_index	dividend_yield
GLD	SPDR Gold Shares	23	Commodities	Precious Metals	Gold	0	0.4	LBMA Gold Price PM	0
IEF	iShares 7-10 Year Treasury Bond ETF	28.5	Fixed Income	US Government Bonds	Intermediate Term Treasury	7.7	0.15	ICE US Treasury 7-10Y Bond Index	3.9
IXN	iShares Global Tech ETF	17.5	Equity	Large Cap	Technology	0	0.39	S&P Global 1200 IT Index	0.24
QQQ	Invesco QQQ Trust	22.1	Equity	Large Cap	Technology	0	0.18	NASDAQ-100 Index	0.43
SPY	State Street SPDR S&P 500 ETF Trust	0	Equity	Large Cap	US Broad Market	0	0.09	S&P 500 Index	1.04
VNQ	Vanguard Real Estate Index Fund ETF Shares	8.9	Real Estate	Large Cap	US REITs	0	0.13	MSCI US IM Real Estate 25/50	0.87

6. T5 Pricing Data Python Script - Screenshot of csv_to_sql_inserts.py running in Terminal showing 2510 total rows written output

The screenshot shows a database management interface with a query editor. The query is as follows:

```

109 # T5-creating a pricing_daily_new table
110 CREATE TABLE pricing_daily_new(
111     Date DATE,
112     Ticker VARCHAR(8), # FK- foreign key from security_masterlist table
113     Open NUMERIC(10,2),
114     High NUMERIC(10,2),
115     Low NUMERIC(10,2),
116     Close NUMERIC(10,2),
117     Adj_Close NUMERIC(10,2),
118     Volume INTEGER
119 ); # closing a pricing_daily_new table
120
121 # inserting values in the pricing_daily_new table
122 # Generated from local CSV files via csv_to_sql_inserts.py
123 # Paste and run this entire file in MySQL Workbench
124
125 # IXN - 502 trading days
126 INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-25','IXN',81.94,82.94,81.69,82.86,81.68,214200);
127 INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-26','IXN',82.88,83.24,82.39,82.88,81.7,122200);
128 INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-27','IXN',83.03,83.5,82.8,82.96,81.77,88400);
  
```

The result grid shows the following data:

Date	Ticker	Open	High	Low	Close	Adj_Close	Volume
2024-06-26	IXN	82.88	83.24	82.39	82.88	81.70	122200
2024-06-27	IXN	83.03	83.50	82.80	82.96	81.77	88400
2024-06-28	IXN	83.41	84.32	82.82	82.82	81.64	212000
2024-07-01	IXN	83.17	83.71	82.05	83.62	82.43	243600
2024-07-02	IXN	82.96	84.15	82.81	84.11	82.91	190300
2024-07-03	IXN	84.11	85.44	84.11	85.41	84.19	121700
2024-07-05	IXN	85.66	86.18	85.53	85.85	84.62	89100
2024-07-08	IXN	86.33	86.75	86.14	86.54	85.30	153900
2024-07-09	IXN	86.96	87.21	86.05	86.45	85.21	134300
2024-07-10	IXN	87.14	87.89	86.82	87.81	86.56	114500
2024-07-11	IXN	88.10	88.10	85.41	85.68	84.46	234400

7. T5 Pricing Table Verification - SELECT Ticker, MIN(Date), MAX(Date) FROM pricing_daily_new GROUP BY Ticker;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema for 'uhnl_1_ca', including tables like 'pricing_daily_new'. The main window shows a SQL query (Query 4) that inserts data into 'pricing_daily_new' and then runs a verification query. The results grid shows the output of the verification query.

```

-- Query 4
-- # closing a pricing_daily_new table
--
-- # inserting values in the pricing_daily_new table
-- # Generated from local CSV files via csv_to_sql_inserts.py
-- # Paste and run this entire file in MySQL Workbench
--
-- SELECT Ticker,
--        MIN(Date),
--        MAX(Date)
-- FROM pricing_daily_new
-- GROUP BY Ticker;
--
-- # IXN - 502 trading days
--
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-25','IXN',81.94,82.94,81.69,82.86,81.68,214200);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-26','IXN',82.88,83.24,82.39,82.88,81.7,122200);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-27','IXN',83.03,83.5,82.8,82.96,81.77,88400);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-06-28','IXN',83.41,84.32,82.82,82.82,81.64,212000);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-07-01','IXN',83.17,83.71,82.05,83.62,82.43,243600);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-07-02','IXN',82.96,84.15,82.81,84.11,82.91,196300);
-- INSERT INTO pricing_daily_new(Date,Ticker,Open,High,Low,Close,Adj_Close,Volume) VALUES ('2024-07-03','IXN',84.11,85.44,84.11,85.41,84.19,121700);
--
-- Result Grid
--
-- Ticker  MIN(Date)  MAX(Date)
-- ----
-- IXN    2024-06-25  2026-06-25
-- OQQ    2024-06-25  2026-06-25
-- IEF    2024-06-25  2026-06-25
-- VNO    2024-06-25  2026-06-25
-- GLD    2024-06-25  2026-06-25

```

8. T6 Benchmark Comparison - CREATE TABLE + INSERT + SELECT * FROM benchmark_comparison;

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema for 'uhnl_1_ca', including tables like 'benchmark_comparison'. The main window shows a SQL query (Query 4) that creates the 'benchmark_comparison' table, inserts data, and then runs a query to select all data from the table. The results grid shows the output of the select query.

```

-- Query 4
-- # T6 - creating benchmark_comparison table
--
-- CREATE TABLE benchmark_comparison(
--   benchmark_id INT PRIMARY KEY,
--   benchmark_name VARCHAR(100), # PK- name of the benchmark
--   ticker VARCHAR(10), # ETF ticker; NULL if blended portfolio
--   annualized_return DECIMAL(9,4), # trailing 24-month total return (%)
--   annualized_risk DECIMAL(9,4), # annualized standard deviation / volatility (%)
--   sharpe_ratio DECIMAL(9,4), # risk-adjusted return over trailing 24 months
--   description VARCHAR(250) # plain English context for client narrative
-- );
-- # closing the benchmark_comparison table
--
-- # inserting value in benchmark_comparison table
--
-- # Source: Yahoo Finance annual returns (2024 + 2025), PortfoliosLab volatility.
-- # 24M annualized return = CAGR of 2024 and 2025 full-year total returns (dividends included).
-- # Sharpe = annualized_return / annualized_risk (consistent with class formula used in stored procs).
--
-- INSERT INTO benchmark_comparison VALUES
-- (1, 'S&P 500', 'SPY', 21.2600, 15.1000, 1.4080,
--  'Broad US equity benchmark - 500 largest US companies. 24M CAGR: 2024 +24.89%, 2025 +17.72%.');
-- (2, '60/40 Portfolio', 'N/A', 13.4400, 9.2000, 1.4600,
--  'Classic UHNW blend - 60% SPY + 40% AGG rebalanced quarterly. 24M blended CAGR. ');
-- (3, 'US Aggregate Bond', 'AGG', 4.2200, 1.2000, 3.5200,
--  'Bloomberg US Aggregate Bond Index - broad US investment-grade fixed income. 24M CAGR: 2024 +1.31%, 2025 +7.19%.');
--
-- Result Grid
--
-- bench... benchmark_name  ticker  annualized_retu...  annualized_risk  sharpe_ratio  description
-- ----
-- 1  S&P 500  SPY  21.2600  15.1000  1.4080  Broad US equity benchmark - 500 largest US companies. 24M CAGR: 2024 +24.89%, 2025 +17.72%.
-- 2  60/40 Portfolio  N/A  13.4400  9.2000  1.4600  Classic UHNW blend - 60% SPY + 40% AGG rebalanced quarterly. 24M blended CAGR.
-- 3  US Aggregate Bond  AGG  4.2200  1.2000  3.5200  Bloomberg US Aggregate Bond Index - broad US investment-grade fixed income. 24M CAGR: 2024 +1.31%, 2025 +7.19%.

```

11.2 Stage 1 - Analysis Query Screenshots

Step 1A - Portfolio Snapshot: SELECT query showing 5 ETFs with weights, market values, duration, expense ratio, dividend yield]

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session No object selected

```

2692
2693
2694 # STEP-1A - portfolio snapshot: shows everything the client currently owns with dollar values and ETF characteristics
2695
2696 * SELECT
2697     ticker,
2698     security_name,
2699     major_asset_class,
2700     portfolio_weight,
2701     ROUND(95000000 * portfolio_weight / 100, 2) AS market_value_usd,
2702     duration_years,
2703     expense_ratio,
2704     dividend_yield,
2705     benchmark_index
2706 FROM uhnl_1_ca.security_masterlist
2707 WHERE ticker IN ('IXN','QQQ','IEF','VNG','GLD')
2708 ORDER BY portfolio_weight DESC;
2709
  
```

Result Grid Filter Rows: Search Export:

ticker	security_name	major_asset_class	portfolio_weight	market_value_usd	duration_years	expense_ratio	dividend_yi...	benchmark_index
IEF	iShares 7-10 Year Treasury Bond ETF	Fixed Income	28.5	27075000	7.7	0.15	3.9	ICE US Treasury 7-10Y Bond Index
GLD	SPDR Gold Shares	Commodities	23	21850000	0	0.4	0	LBMA Gold Price PM
QQQ	Invesco QQQ Trust	Equity	22.1	20995000	0	0.18	0.43	NASDAQ-100 Index
IXN	iShares Global Tech ETF	Equity	17.5	16625000	0	0.39	0.24	S&P Global 1200 IT Index
VNG	Vanguard Real Estate Index Fund ETF Shares	Real Estate	8.9	8455000	0	0.13	3.87	MSCI US RE Real Estate 25/50

Step 1B - Return Analysis (12 Months call from stored procedure): 5 tickers ordered by Sharpe DESC - IXN top (0.11972), IEF bottom (0.046679)

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - Views
 - Stored Procedures
 - Functions

Object Info Session Schema: uhnl_1_ca

```

2710 # STEP 1B - stored procedure to calculate return, risk and sharpe for each ticker
2711 # call with 12, 18, or 24 to get results for that time window
2712
2713 DELIMITER $$
2714 * CREATE PROCEDURE sp_return_analysis(IN months_back INT)
2715 BEGIN
2716     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2717
2718     SELECT
2719         y.ticker,
2720         ROUND(AVG(y.ror), 8) AS avg_daily_return,
2721         ROUND(STD(y.ror), 8) AS daily_risk,
2722         ROUND(AVG(y.ror) * 252, 6) AS annualized_return,
2723         ROUND(STD(y.ror) * SQRT(252), 6) AS annualized_risk,
2724         ROUND(AVG(y.ror) / STD(y.ror), 6) AS sharpe_ratio
2725     FROM (
2726         SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2727         FROM (
2728             SELECT Ticker AS ticker, Adj_Close AS p1,
2729                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2730             FROM uhnl_1_ca.pricing_daily_new
2731             WHERE Date >= @start_date
2732                   AND Ticker IN ('IXN','QQQ','IEF','VNG','GLD')
2733         ) z
2734     ) y
2735     GROUP BY y.ticker
2736     ORDER BY sharpe_ratio DESC;
2737
2738 DELIMITER ;
2739 * # call for 12 months
2740 CALL sp_return_analysis(12);
  
```

Result Grid Filter Rows: Search Export:

ticker	avg_daily_retu...	daily_risk	annualized_return	annualized_risk	sharpe_ratio
IXN	0.00187262	0.01588186	0.471900	0.252117	0.117909
QQQ	0.00114214	0.01130482	0.287819	0.179458	0.101031
VNG	0.00050923	0.00857625	0.133367	0.136144	0.061709
GLD	0.00065539	0.01747714	0.240759	0.277441	0.054665
IEF	0.00013999	0.00298588	0.034521	0.047399	0.045879

Step 1B - Return Analysis (18 Months call from stored procedure): 5 tickers - IXN top Sharpe (0.080834), VNQ bottom (0.039036)

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - Views
 - Stored Procedures
 - Functions

Object Info Session Schema: uhnl_1_ca

```

2710 # STEP 1B - stored procedure to calculate return, risk and sharpe for each ticker
2711 # call with 12, 18, or 24 to get results for that time window
2712
2713 DELIMITER $$
2714 CREATE PROCEDURE sp_return_analysis(IN months_back INT)
2715 BEGIN
2716     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2717     SELECT
2718         y.ticker,
2719         ROUND(AVG(y.ror), 8) AS avg_daily_return,
2720         ROUND(STD(y.ror), 8) AS daily_risk,
2721         ROUND(AVG(y.ror) * 252, 6) AS annualized_return,
2722         ROUND(STD(y.ror) * SORT(252), 6) AS annualized_risk,
2723         ROUND(AVG(y.ror) / STD(y.ror), 6) AS sharpe_ratio
2724     FROM (
2725         SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2726         FROM (
2727             SELECT Ticker AS ticker, Adj_Close AS p1,
2728                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2729             FROM uhnl_1_ca.pricing_daily_new
2730             WHERE Date >= @start_date
2731                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')
2732         ) z
2733     ) y
2734     GROUP BY y.ticker
2735     ORDER BY sharpe_ratio DESC;
2736 END$$
2737 DELIMITER ;
2738 # call for 12 months
2739 CALL sp_return_analysis(12);
  
```

Result Grid

ticker	avg_daily_return	daily_risk	annualized_return	annualized_risk	sharpe_ratio
IXN	0.00148868	0.01800318	0.375147	0.285782	0.08269
GLD	0.00128095	0.01601223	0.317759	0.254186	0.076749
IEF	0.00023185	0.00342627	0.058425	0.054422	0.067628
QQQ	0.00095829	0.01430252	0.241480	0.227046	0.067002
VNQ	0.00042789	0.01024147	0.107828	0.162578	0.04178

Step 1B - Return Analysis (24 Months call from stored procedure): 5 tickers - GLD top Sharpe (0.080797), IEF bottom (0.045898)

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - Views
 - Stored Procedures
 - Functions

Object Info Session Schema: uhnl_1_ca

```

2710 # STEP 1B - stored procedure to calculate return, risk and sharpe for each ticker
2711 # call with 12, 18, or 24 to get results for that time window
2712
2713 DELIMITER $$
2714 CREATE PROCEDURE sp_return_analysis(IN months_back INT)
2715 BEGIN
2716     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2717     SELECT
2718         y.ticker,
2719         ROUND(AVG(y.ror), 8) AS avg_daily_return,
2720         ROUND(STD(y.ror), 8) AS daily_risk,
2721         ROUND(AVG(y.ror) * 252, 6) AS annualized_return,
2722         ROUND(STD(y.ror) * SORT(252), 6) AS annualized_risk,
2723         ROUND(AVG(y.ror) / STD(y.ror), 6) AS sharpe_ratio
2724     FROM (
2725         SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2726         FROM (
2727             SELECT Ticker AS ticker, Adj_Close AS p1,
2728                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2729             FROM uhnl_1_ca.pricing_daily_new
2730             WHERE Date >= @start_date
2731                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNQ', 'GLD')
2732         ) z
2733     ) y
2734     GROUP BY y.ticker
2735     ORDER BY sharpe_ratio DESC;
2736 END$$
2737 DELIMITER ;
2738 # call for 24 months
2739 CALL sp_return_analysis(24);
  
```

Result Grid

ticker	avg_daily_return	daily_risk	annualized_return	annualized_risk	sharpe_ratio
GLD	0.00118432	0.01466641	0.298448	0.232822	0.08075
IXN	0.00122836	0.0173436	0.309546	0.275321	0.070825
QQQ	0.00090480	0.01380263	0.228009	0.221015	0.044988
VNQ	0.00049824	0.01011153	0.125808	0.160516	0.049373
IEF	0.00017329	0.00359252	0.043669	0.057029	0.048237

Step 1C - Weighted Return and Risk per Ticker: GLD highest weighted return (0.02722736), IEF lowest (0.00470569)- (24 Months call from stored procedure):

Administration Schemas Query 4 A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

boston_capital

sys

uhni_1_ca

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: uhni_1_ca

```

DELIMITER $$
2751 CREATE PROCEDURE sp_weighted_analysis(IN months_back INT)
2752 BEGIN
2753 SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2754 SELECT
2755     y.ticker,
2756     sm.security_name,
2757     sm.major_asset_class,
2758     sm.portfolio_weight,
2759     ROUND(AVG(y.ror), 8) AS avg_daily_return,
2760     ROUND(STD(y.ror), 8) AS daily_risk,
2761     ROUND(AVG(y.ror) / STD(y.ror), 8) AS sharpe_ratio,
2762     ROUND(sm.portfolio_weight * AVG(y.ror), 8) AS weighted_return,
2763     ROUND(sm.portfolio_weight * STD(y.ror), 8) AS weighted_risk
2764 FROM (
2765     SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2766     FROM (
2767         SELECT Ticker AS ticker, Adj_Close AS p1,
2768             LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2769         FROM uhni_1_ca.pricing_daily_new
2770         WHERE Date >= @start_date
2771             AND Ticker IN ('DXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2772     ) z
2773 ) y
2774 JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2775 GROUP BY y.ticker, sm.security_name, sm.major_asset_class, sm.portfolio_weight
2776 ORDER BY weighted_return DESC;
2777 END$$
2778 DELIMITER ;
2779 CALL sp_weighted_analysis(24);
2780

```

Result Grid

Filter Rows: Search Export:

ticker	security_name	major_asset_class	portfolio_weight	avg_daily_retu...	daily_risk	sharpe_ratio	weighted_ret...	weighted_risk
GLD	SPDR Gold Shares	Commodities	23	0.00118432	0.01466641	0.08075	0.027223928	0.33732748
DXN	iShares Global Tech ETF	Equity	17.5	0.00122836	0.0173436	0.070825	0.02148624	0.30351288
QQQ	Invesco QQQ Trust	Equity	22.1	0.00090480	0.01392263	0.064988	0.01999603	0.30769022
IEF	iShares 7-10 Year Treasury Bond ETF	Fixed Income	28.5	0.00017329	0.00359252	0.048237	0.00493878	0.10038678
VNO	Vanguard Real Estate Index Fund ETF Shares	Real Estate	8.9	0.00049924	0.01011153	0.048373	0.0044432	0.08996265

Step 1D(0) - Client Risk Profiling: avg_portfolio_return=0.00079908, avg_portfolio_risk=0.0119108, Moderate Risk Tolerance (24 Months call from stored procedure):

Administration Schemas Query 4 A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

boston_capital

sys

uhni_1_ca

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: uhni_1_ca

```

4823
2824 # STEP 1D(0) - stored procedure to calculate average portfolio return, risk and client risk profile
2825 # call with 24 to get results for that time window
2826
2827 DELIMITER $$
2828
2829 CREATE PROCEDURE sp_risk_profiling(IN months_back INT)
2830 BEGIN
2831 SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2832
2833 SELECT
2834     ROUND(AVG(expected_return), 8) AS avg_portfolio_return,
2835     ROUND(AVG(risk), 8) AS avg_portfolio_risk,
2836     CASE
2837         WHEN AVG(risk) > 0.02 THEN 'High Risk Tolerance'
2838         WHEN AVG(risk) BETWEEN 0.01
2839             AND 0.02 THEN 'Moderate Risk Tolerance'
2840         ELSE 'Low Risk Tolerance'
2841     END AS risk_profile
2842 FROM (
2843     SELECT
2844         y.ticker,
2845         AVG(y.ror) AS expected_return,
2846         STD(y.ror) AS risk
2847     FROM (
2848         SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2849         FROM (
2850             SELECT Ticker AS ticker, Adj_Close AS p1,
2851                 LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2852             FROM uhni_1_ca.pricing_daily_new
2853             WHERE Date >= @start_date
2854                 AND Ticker IN ('DXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2855         ) z
2856     ) y
2857     JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2858     GROUP BY y.ticker, sm.security_name, sm.major_asset_class, sm.portfolio_weight
2859 )

```

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- Filter objects
- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - Views
 - Stored Procedures
 - Functions

Object Info Session Schema: uhnl_1_ca

```

2853 SELECT Ticker AS ticker, Adj_Close AS p1,
2854        LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2855 FROM uhnl_1_ca.pricing_daily_new
2856 WHERE Date >= @start_date
2857        AND Ticker IN ('IXN','QOO','IEF','VNO','GLD')
2858 ) z
2859 ) y
2860 GROUP BY y.ticker
2861 ) risk_table;
2862
2863 END$$
2864
2865 DELIMITER ;
2866
2867 # call
2868 CALL sp_risk_profiling(24);
2869
100% 1:2869 1 error found

```

Result Grid

avg_portfolio_return	avg_portfolio_risk	risk_profile
0.00079800	0.01192734	Moderate Risk Tolerance

Step 1D - Total Portfolio Return, Risk and Sharpe: 0.07804552, 1.13939773, 0.068497 (12 Months call from stored procedure):

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

- Filter objects
- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - Views
 - Stored Procedures
 - Functions

Object Info Session Schema: uhnl_1_ca

```

2781 # STEP 1D - stored procedure to calculate total portfolio level return, risk and sharpe ratio
2782 # call with 12, 18, or 24 to get results for that time window
2783 DELIMITER $$
2784
2785 CREATE PROCEDURE sp_portfolio_summary(IN months_back INT)
2786 BEGIN
2787     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2788
2789     SELECT
2790         ROUND(SUM(weighted_return), 8) AS total_portfolio_return,
2791         ROUND(SUM(weighted_risk), 8) AS total_portfolio_risk,
2792         ROUND(SUM(weighted_return) / SUM(weighted_risk), 6) AS portfolio_sharpe
2793     FROM (
2794         SELECT
2795             sm.portfolio_weight * AVG(y.ror) AS weighted_return,
2796             sm.portfolio_weight * STD(y.ror) AS weighted_risk
2797         FROM (
2798             SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2799             FROM (
2800                 SELECT Ticker AS ticker, Adj_Close AS p1,
2801                      LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2802                 FROM uhnl_1_ca.pricing_daily_new
2803                 WHERE Date >= @start_date
2804                      AND Ticker IN ('IXN','QOO','IEF','VNO','GLD')
2805             ) z
2806             JOIN uhnl_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2807             GROUP BY sm.ticker, sm.portfolio_weight
2808         ) combined;
2809     END$$
2810 DELIMITER ;
2811
2812 # call for 12 months
2813 CALL sp_portfolio_summary(12);
2814
100% 1:2811 1 error found

```

Result Grid

total_portfolio_return	total_portfolio_risk	portfolio_sharpe
0.08860051	1.09118941	0.081198

Step 1D - Total Portfolio Return, Risk and Sharpe: 0.07804552, 1.13939773, 0.068497 (18 Months call from stored procedure):

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Schemas' tree with 'uhni_1_ca' selected. The right pane shows the 'Query 4' window with the following SQL code:

```

2782 # STEP 1D - stored procedure to calculate total portfolio level return, risk and sharpe ratio
2783 # call with 12, 18, or 24 to get results for that time window
2784 DELIMITER $$
2785 CREATE PROCEDURE sp_portfolio_summary(IN months_back INT)
2786 BEGIN
2787     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2788     SELECT
2789         ROUND(SUM(weighted_return), 8) AS total_portfolio_return,
2790         ROUND(SUM(weighted_risk), 8) AS total_portfolio_risk,
2791         ROUND(SUM(weighted_return) / SUM(weighted_risk), 6) AS portfolio_sharpe
2792     FROM (
2793         SELECT
2794             sm.portfolio_weight * AVG(y.ror) AS weighted_return,
2795             sm.portfolio_weight * STD(y.ror) AS weighted_risk
2796         FROM (
2797             SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2798             FROM (
2799                 SELECT Ticker AS ticker, Adj_Close AS p1,
2800                     LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2801                 FROM uhni_1_ca.pricing_daily_new
2802                 WHERE Date >= @start_date
2803                     AND Ticker IN ('EXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2804             ) z
2805             ) y
2806         JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2807         GROUP BY sm.ticker, sm.portfolio_weight
2808     ) combined;
2809 END$$
2810 DELIMITER ;
2811
2812 # call for 12 months
2813 CALL sp_portfolio_summary(18);

```

The bottom of the window shows the 'Result Grid' with the following data:

total_portfolio_return	total_portfolio_risk	portfolio_sharpe
0.08694783	1.18827745	0.072919

Step 1D - Total Portfolio Return, Risk and Sharpe: 0.07804552, 1.13939773, 0.068497 (24 Months call from stored procedure):

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Schemas' tree with 'uhni_1_ca' selected. The right pane shows the 'Query 4' window with the following SQL code:

```

2782 # STEP 1D - stored procedure to calculate total portfolio level return, risk and sharpe ratio
2783 # call with 12, 18, or 24 to get results for that time window
2784 DELIMITER $$
2785 CREATE PROCEDURE sp_portfolio_summary(IN months_back INT)
2786 BEGIN
2787     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2788     SELECT
2789         ROUND(SUM(weighted_return), 8) AS total_portfolio_return,
2790         ROUND(SUM(weighted_risk), 8) AS total_portfolio_risk,
2791         ROUND(SUM(weighted_return) / SUM(weighted_risk), 6) AS portfolio_sharpe
2792     FROM (
2793         SELECT
2794             sm.portfolio_weight * AVG(y.ror) AS weighted_return,
2795             sm.portfolio_weight * STD(y.ror) AS weighted_risk
2796         FROM (
2797             SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2798             FROM (
2799                 SELECT Ticker AS ticker, Adj_Close AS p1,
2800                     LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2801                 FROM uhni_1_ca.pricing_daily_new
2802                 WHERE Date >= @start_date
2803                     AND Ticker IN ('EXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2804             ) z
2805             ) y
2806         JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2807         GROUP BY sm.ticker, sm.portfolio_weight
2808     ) combined;
2809 END$$
2810 DELIMITER ;
2811
2812 # call for 12 months
2813 CALL sp_portfolio_summary(24);

```

The bottom of the window shows the 'Result Grid' with the following data:

total_portfolio_return	total_portfolio_risk	portfolio_sharpe
0.07811354	1.14091011	0.068466

Step 1E - Correlation Matrix: 10 pairs - IXN/QQQ=0.9616 at top, IEF/QQQ=-0.0222 at bottom 12 Months call from stored procedure):

Administration Schemas Query 4 # AI Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

boston_capital

sys

uhni_1_ca

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: uhni_1_ca

```

2864
2865 # STEP 1E - stored procedure to calculate pairwise correlation between all tickers
2866 # call with 12, 18, or 24 to get results for that time window
2867
2868 DELIMITER $$
2869
2870 CREATE PROCEDURE sp_correlation_matrix(IN months_back INT)
2871 BEGIN
2872     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2873
2874     SELECT
2875         a.ticker AS ticker_1,
2876         b.ticker AS ticker_2,
2877         ROUND(
2878             (AVG(a.ror * b.ror) - AVG(a.ror) * AVG(b.ror)) /
2879             (STD(a.ror) * STD(b.ror))
2880         , 4) AS correlation
2881     FROM (
2882         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2883         FROM (
2884             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2885                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2886             FROM uhni_1_ca.pricing_daily_new
2887             WHERE Date >= @start_date
2888                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2889         ) z
2890     ) a
2891     JOIN (
2892         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2893         FROM (
2894             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2895                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2896             FROM uhni_1_ca.pricing_daily_new
2897
2898             WHERE Date >= @start_date
2899                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2900         ) z
2901     ) b ON a.date = b.date AND a.ticker < b.ticker
2902     GROUP BY a.ticker, b.ticker
2903     ORDER BY correlation DESC;
2904
2905 END$$
2906
2907 DELIMITER ;
2908
2909 # call for 12 months
2910 CALL sp_correlation_matrix(12);
2911
100% 1:2910 1 error found

```

Result Grid Filter Rows Search Export

ticker_1	ticker_2	correlation
IXN	QQQ	0.9625
IEF	VNO	0.4202
GLD	QQQ	0.3111
GLD	IXN	0.3074
GLD	IEF	0.2303
IEF	QQQ	0.1711
IEF	IXN	0.156
QQQ	VNO	0.1495
GLD	VNO	0.1464
IXN	VNO	0.0619

Object Info Session

Schema: uhni_1_ca

Result Grid

Form Editor

Field Types

Export

Step 1E - Correlation Matrix: 10 pairs - IXN/QQQ=0.9616 at top, IEF/QQQ=-0.0222 at bottom 18 Months call from stored procedure):

Administration Schemas Query 4 # AI Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

boston_capital

sys

uhnl_1_ca

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: uhn1_1_ca

```

2863
2864
2865 # STEP 1E - stored procedure to calculate pairwise correlation between all tickers
2866 # call with 12, 18, or 24 to get results for that time window
2867
2868 DELIMITER $$
2869
2870 CREATE PROCEDURE sp_correlation_matrix(IN months_back INT)
2871 BEGIN
2872     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2873
2874     SELECT
2875         a.ticker AS ticker_1,
2876         b.ticker AS ticker_2,
2877         ROUND(
2878             (AVG(a.ror * b.ror) - AVG(a.ror) * AVG(b.ror)) /
2879             (STD(a.ror) * STD(b.ror))
2880         , 4) AS correlation
2881     FROM (
2882         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2883         FROM (
2884             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2885                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2886             FROM uhn1_1_ca.pricing_daily_new
2887             WHERE Date >= @start_date
2888                   AND Ticker IN ('IXN','QQQ','IEF','VNO','GLD')
2889         ) z
2890     ) a
2891     JOIN (
2892         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2893         FROM (
2894             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2895                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2896             FROM uhn1_1_ca.pricing_daily_new
2897
2898             WHERE Date >= @start_date
2899                   AND Ticker IN ('IXN','QQQ','IEF','VNO','GLD')
2900         ) z
2901     ) b ON a.date = b.date AND a.ticker < b.ticker
2902     GROUP BY a.ticker, b.ticker
2903     ORDER BY correlation DESC;
2904 END$$
2905
2906 DELIMITER ;
2907
2908 # call for 12 months
2909 CALL sp_correlation_matrix(12);
2910
2911
100% 32/2909 1 error found

```

Result Grid

Filter Rows: Search Export:

ticker_1	ticker_2	correlation
IXN	QQQ	0.9618
QQQ	VNO	0.4135
IXN	VNO	0.322
IEF	VNO	0.2873
GLD	IXN	0.21
GLD	QQQ	0.1772
GLD	IEF	0.1598
GLD	VNO	0.1119
IEF	IXN	0.0037
IEF	QQQ	-0.0052

Object Info Session

Schema: uhn1_1_ca

Result Grid

Form Editor

Field Types

Step 1E - Correlation Matrix: 10 pairs - IXN/QQQ=0.9616 at top, IEF/QQQ=-0.0222 at bottom 24 Months call from stored procedure):

Administration Schemas Query 4 AI Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

boston_capital

sys

uhni_1_ca

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: uhni_1_ca

```

2863
2864
2865 # STEP 1E - stored procedure to calculate pairwise correlation between all tickers
2866 # call with 12, 18, or 24 to get results for that time window
2867
2868 DELIMITER $$
2869
2870 CREATE PROCEDURE sp_correlation_matrix(IN months_back INT)
2871 BEGIN
2872     SET @start_date = DATE_SUB(CURDATE(), INTERVAL months_back MONTH);
2873
2874     SELECT
2875         a.ticker AS ticker_1,
2876         b.ticker AS ticker_2,
2877         ROUND(
2878             (AVG(a.ror * b.ror) - AVG(a.ror) * AVG(b.ror)) /
2879             (STD(a.ror) * STD(b.ror))
2880         , 4) AS correlation
2881     FROM (
2882         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2883         FROM (
2884             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2885                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2886             FROM uhni_1_ca.pricing_daily_new
2887             WHERE Date >= @start_date
2888                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2889         ) z
2890     ) a
2891     JOIN (
2892         SELECT z.ticker, z.date, (z.p1 - z.p0) / z.p0 AS ror
2893         FROM (
2894             SELECT Ticker AS ticker, Date AS date, Adj_Close AS p1,
2895                    LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2896             FROM uhni_1_ca.pricing_daily_new
2897
2898             WHERE Date >= @start_date
2899                   AND Ticker IN ('IXN', 'QQQ', 'IEF', 'VNO', 'GLD')
2900         ) z
2901     ) b ON a.date = b.date AND a.ticker < b.ticker
2902     GROUP BY a.ticker, b.ticker
2903     ORDER BY correlation DESC;
2904
2905 END$$
2906
2907 DELIMITER ;
2908
2909 # call for 12 months
2910 CALL sp_correlation_matrix(24);
2911
100% 32:2909 1 error found

```

Result Grid

ticker_1	ticker_2	correlation
IXN	QQQ	0.9616
QQQ	VNO	0.3774
IEF	VNO	0.3255
IXN	VNO	0.2986
GLD	IXN	0.2209
GLD	QQQ	0.1823
GLD	IEF	0.1826
GLD	VNO	0.1431
IEF	IXN	-0.0155
IEF	QQQ	-0.0221

Object Info Session

Schema: uhni_1_ca

Result Grid

Form Editor

Field Types

Step 1F - Benchmark Comparison: US Aggregate Bond 3.52, 60/40 1.46, S&P500 1.408, Client Portfolio 1.0874]

Administration Schemas Query 4 A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

# STEP-1F - compares client portfolio return, risk and sharpe against standard market benchmarks

2894 SELECT
2895 'Client Portfolio' AS entity,
2896 ROUND(SUM(weighted_return) * 252 * 100, 4) AS annualized_return,
2897 ROUND(SUM(weighted_risk) * SQRT(252) * 100, 4) AS annualized_risk,
2898 ROUND(SUM(weighted_return) * 252 / (SUM(weighted_risk) * SQRT(252)), 4) AS sharpe_ratio
2899 FROM (
2900 SELECT
2901 (sm.portfolio_weight / 100) * AVG(y.ror) AS weighted_return,
2902 (sm.portfolio_weight / 100) * STD(y.ror) AS weighted_risk
2903 FROM (
2904 SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
2905 FROM (
2906 SELECT Ticker AS ticker, Adj_Close AS p1,
2907 LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
2908 FROM uhni_1_ca.pricing_daily_new
2909 WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
2910 AND Ticker IN ('IXN','000','IEF','VNO','GLD')
2911 ) z
2912 ) y
2913 JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
2914 GROUP BY sm.ticker, sm.portfolio_weight
2915 ) combined UNION ALL
2916 SELECT
2917 benchmark_name AS entity,
2918 annualized_return,
2919 annualized_risk,
2920 sharpe_ratio
2921 FROM uhni_1_ca.benchmark_comparison
2922 ORDER BY sharpe_ratio DESC;
  
```

100% 1:2910

Result Grid Filter Rows: Search Export:

entity	annualized_return	annualized_risk	sharpe_ratio
US Aggregate Bond	4.22	1.2	3.52
60/40 Portfolio	13.44	9.2	1.46
S&P 500	21.26	15.1	1.408
Client Portfolio	19.6675	18.0874	1.0874

Result Grid

11.3 Stage 2 - Optimization Query Screenshots

Step 2A - Per Ticker Before vs After: CASE statement with new weights showing old/new weighted return and risk side by side]

Administration Schemas Query 4 # A1 Assignment* Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhnl_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session
No object selected

```

# Stage 2- Optimizing portfolio

# STEP-2A - shows each ticker with old and new weights, return, risk and sharpe to compare before and after rebalancing

SELECT
  sm.ticker,
  sm.security_name,
  sm.major_asset_class,
  sm.portfolio_weight
  AS old_weight,
  CASE sm.ticker
    WHEN 'IXN' THEN 22.0
    WHEN 'QQQ' THEN 33.0
    WHEN 'IEF' THEN 10.0
    WHEN 'VNO' THEN 5.0
    WHEN 'GLD' THEN 30.0
    ELSE 0
  END
  AS new_weight,
  ROUND(AVG(y.ror), 8)
  AS avg_daily_return,
  ROUND(STD(y.ror), 8)
  AS daily_risk,
  ROUND(AVG(y.ror) / STD(y.ror), 6)
  AS sharpe_ratio,
  ROUND(sm.portfolio_weight * AVG(y.ror), 8)
  AS old_weighted_return,
  ROUND(CASE sm.ticker
    WHEN 'IXN' THEN 22.0
    WHEN 'QQQ' THEN 33.0
    WHEN 'IEF' THEN 10.0
    WHEN 'VNO' THEN 5.0
    WHEN 'GLD' THEN 30.0
    ELSE 0
  END * AVG(y.ror), 8)
  AS new_weighted_return,
  ROUND(sm.portfolio_weight * STD(y.ror), 8)
  AS old_weighted_risk,
  ROUND(CASE sm.ticker
    WHEN 'IXN' THEN 22.0
    WHEN 'QQQ' THEN 33.0
    WHEN 'IEF' THEN 10.0
    WHEN 'VNO' THEN 5.0
    WHEN 'GLD' THEN 30.0
    ELSE 0
  END * STD(y.ror), 8)
  AS new_weighted_risk
FROM (
  SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
  FROM (
    SELECT Ticker AS ticker, Adj_Close AS p1,
      LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
    FROM uhnl_1_ca.pricing_daily_new
    WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
    AND Ticker IN ('IXN','QQQ','IEF','VNO','GLD')
  ) z
  ) y
JOIN uhnl_1_ca.security_masterlist sm ON y.ticker = sm.ticker
GROUP BY sm.ticker, sm.security_name, sm.major_asset_class, sm.portfolio_weight
ORDER BY new_weight DESC;
  
```

Result Grid Filter Rows: Search Export:

ticker	security_name	major_asset_class	old_weight	new_weight	avg_daily_retu...	daily_risk	sharpe_ratio	old_weighted_ret...	new_weighted_return	old_weighted_risk	new_weighted_risk
QQQ	Invesco QQQ Trust	Equity	22.1	33.0	0.00091047	0.01389512	0.065525	0.02012145	0.03004561	0.30708206	0.45853882
GLD	SPDR Gold Shares	Commodities	23	30.0	0.00118380	0.01465152	0.080797	0.02722736	0.03551395	0.336985	0.43954565
IXN	iShares Global Tech ETF	Equity	17.5	22.0	0.00122565	0.01730901	0.07081	0.02144882	0.02696436	0.30290767	0.38079822
IEF	iShares 7-10 Year Treasury Bond ETF	Fixed Income	28.5	10.0	0.00018511	0.00359734	0.045898	0.00470569	0.00165112	0.10252406	0.03597336
VNO	Vanguard Real Estate Index Fund ETF Shares	Real Estate	8.9	5.0	0.00051035	0.010101	0.050524	0.00454209	0.00255174	0.08898983	0.05050502

Object Info Session
No object selected

Step 2B - Portfolio Level Before vs After: Before Sharpe 0.068497 vs After Sharpe 0.070843]

Administration Schemas Query 4 A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

-- STEP-2B - compares total portfolio return, risk and sharpe before and after rebalancing in one single output

SELECT
  scenario,
  ROUND(SUM(weighted_return), 8) AS total_portfolio_return,
  ROUND(SUM(weighted_risk), 8) AS total_portfolio_risk,
  ROUND(SUM(weighted_return) / SUM(weighted_risk), 8) AS portfolio_sharpe
FROM (
  SELECT
    'Before Rebalancing' AS scenario,
    sm.ticker,
    sm.portfolio_weight * AVG(y.ror) AS weighted_return,
    sm.portfolio_weight * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
             LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
            AND Ticker IN ('IXN', '000', 'IEF', 'VNO', 'GLD')
    ) z
  ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) UNION ALL
SELECT
  'After Rebalancing - Scenario C Growth (CHOSEN)' AS scenario,
  sm.ticker,
  CASE sm.ticker
    WHEN 'IXN' THEN 22.0
    WHEN '000' THEN 33.0
    WHEN 'IEF' THEN 10.0
    WHEN 'VNO' THEN 5.0
    WHEN 'GLD' THEN 30.0
    ELSE 0
  END * AVG(y.ror) AS weighted_return,
  CASE sm.ticker
    WHEN 'IXN' THEN 22.0
    WHEN '000' THEN 33.0
    WHEN 'IEF' THEN 10.0
    WHEN 'VNO' THEN 5.0
    WHEN 'GLD' THEN 30.0
    ELSE 0
  END * STD(y.ror) AS weighted_risk
FROM (
  SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
  FROM (
    SELECT Ticker AS ticker, Adj_Close AS p1,
           LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
    FROM uhni_1_ca.pricing_daily_new
    WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
          AND Ticker IN ('IXN', '000', 'IEF', 'VNO', 'GLD')
    ) z
  ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) combined
GROUP BY scenario ORDER BY scenario DESC;

```

Result Grid

scenario	total_portfolio_return	total_portfolio_risk	portfolio_sharpe
Before Rebalancing	0.07804552	1.13839773	0.068497
After Rebalancing - Scenario C Growth (CHOSEN)	0.09672678	1.36536107	0.070843

Step 2C - Three Scenario Analysis: Scenario C top (0.070843), Scenario A second (0.0704), Scenario B third (0.070048)]

Administration Schemas Query 4 AI Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

# STEP-2C - three optimized scenarios all designed to beat current portfolio sharpe of 0.0685

SELECT
  scenario,
  ROUND(SUM(weighted_return), 0) AS total_portfolio_return,
  ROUND(SUM(weighted_risk), 0) AS total_portfolio_risk,
  ROUND(SUM(weighted_return) / SUM(weighted_risk), 0) AS portfolio_sharpe
FROM (
  SELECT
    "Scenario A - Conservative (IXN 10, 000 28, IEF 17, VNQ 10, GLD 35)" AS scenario,
    sm.ticker,
    CASE sm.ticker
      WHEN 'IXN' THEN 10.0
      WHEN '000' THEN 28.0
      WHEN 'IEF' THEN 17.0
      WHEN 'VNQ' THEN 10.0
      WHEN 'GLD' THEN 35.0
      ELSE 0
    END * AVG(y.ror) AS weighted_return,
    CASE sm.ticker
      WHEN 'IXN' THEN 10.0
      WHEN '000' THEN 28.0
      WHEN 'IEF' THEN 17.0
      WHEN 'VNQ' THEN 10.0
      WHEN 'GLD' THEN 35.0
      ELSE 0
    END * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
        LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
      AND Ticker IN ('IXN', '000', 'IEF', 'VNQ', 'GLD')
    ) z
    JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
    GROUP BY sm.ticker, sm.portfolio_weight
  ) y
  UNION ALL
  SELECT
    "Scenario B - Balanced Optimized [RECOMMENDED] (IXN 8, 000 32, IEF 12, VNQ 13, GLD 35)" AS scenario,
    sm.ticker,
    CASE sm.ticker
      WHEN 'IXN' THEN 8.0
      WHEN '000' THEN 32.0
      WHEN 'IEF' THEN 12.0
      WHEN 'VNQ' THEN 13.0
      WHEN 'GLD' THEN 35.0
      ELSE 0
    END * AVG(y.ror) AS weighted_return,
    CASE sm.ticker
      WHEN 'IXN' THEN 8.0
      WHEN '000' THEN 32.0
      WHEN 'IEF' THEN 12.0
      WHEN 'VNQ' THEN 13.0
      WHEN 'GLD' THEN 35.0
      ELSE 0
    END * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
  )

```

Administration Schemas Query 4 AI Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

Administration Schemas Query 4 A1 Assignment* Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

3616 FROM (
3617     SELECT Ticker AS ticker, Adj_Close AS p1,
3618            LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
3619     FROM uhni_1_ca.pricing_daily_new
3620     WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
3621           AND Ticker IN ('IXN','QQQ','IEF','VNO','GLD')
3622 ) z
3623 ) y
3624 JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
3625 GROUP BY sm.ticker, sm.portfolio_weight
3626
3627 UNION ALL
3628
3629 SELECT
3630     'Scenario C - Growth (IXN 22, QQQ 33, IEF 10, VNO 5, GLD 30)' AS scenario,
3631     sm.ticker,
3632     CASE sm.ticker
3633     WHEN 'IXN' THEN 22.0
3634     WHEN 'QQQ' THEN 33.0
3635     WHEN 'IEF' THEN 10.0
3636     WHEN 'VNO' THEN 5.0
3637     WHEN 'GLD' THEN 30.0
3638     ELSE 0
3639     END * AVG(y.ror) AS weighted_return,
3640     CASE sm.ticker
3641     WHEN 'IXN' THEN 22.0
3642     WHEN 'QQQ' THEN 33.0
3643     WHEN 'IEF' THEN 10.0
3644     WHEN 'VNO' THEN 5.0
3645     WHEN 'GLD' THEN 30.0
3646     ELSE 0
3647     END * STD(y.ror) AS weighted_risk
3648 FROM (
3649     SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
3650     FROM (
3651         SELECT Ticker AS ticker, Adj_Close AS p1,
3652                LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
3653         FROM uhni_1_ca.pricing_daily_new
3654         WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
3655               AND Ticker IN ('IXN','QQQ','IEF','VNO','GLD')
3656     ) z
3657     ) y
3658 JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
3659 GROUP BY sm.ticker, sm.portfolio_weight
3660 ) combined
3661 GROUP BY scenario
3662 ORDER BY portfolio_sharpe DESC;
3663
3664 100% 48:3615

```

Result Grid Filter Rows: Search Export:

scenario	total_portfolio_return	total_portfolio_risk	portfolio_sharpe
Scenario C - Growth (IXN 22, QQQ 33, IEF 10, VNO 5, GLD 30)	0.09672678	1.36536107	0.070843
Scenario A - Conservative (IXN 10, QQQ 28, IEF 17, VNO 10, GLD 35)	0.08709309	1.23712134	0.0704
Scenario B - Balanced Optimized [RECOMMENDED] (IXN 8, QQQ 32, IEF 12, VNO 13, GLD 35)	0.08898916	1.27040012	0.070048

Result Grid Form Editor

Step 2D - Final Benchmark Comparison: Optimized Portfolio Sharpe 1.1246 vs Original 1.0874 vs benchmarks]

Administration Schemas Query 4 AI Assignment* Limit to 1000 rows

SCHEMAS

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session
No object selected

```

-- STEP-2D - compares original portfolio, optimized portfolio and all three benchmarks in one single output
SELECT
  'Original Portfolio' AS entity,
  ROUND(SUM(weighted_return) * 252 * 100, 4) AS annualized_return,
  ROUND(SUM(weighted_risk) * SQRT(252) * 100, 4) AS annualized_risk,
  ROUND(SUM(weighted_return) * 252 / (SUM(weighted_risk) * SQRT(252)), 4) AS sharpe_ratio
FROM (
  SELECT
    (sm.portfolio_weight / 100) * AVG(y.ror) AS weighted_return,
    (sm.portfolio_weight / 100) * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
        LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
      AND Ticker IN ('IXN','000','IEF','VNO','GLD')
    ) z
    ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) combined
UNION ALL

```

3691 SELECT

```

  'Optimized Portfolio - Scenario C Growth (CHOSEN)' AS entity,
  ROUND(SUM(weighted_return) * 252 * 100, 4) AS annualized_return,
  ROUND(SUM(weighted_risk) * SQRT(252) * 100, 4) AS annualized_risk,
  ROUND(SUM(weighted_return) * 252 / (SUM(weighted_risk) * SQRT(252)), 4) AS sharpe_ratio
FROM (
  SELECT
    (CASE sm.ticker
      WHEN 'IXN' THEN 22.0
      WHEN '000' THEN 33.0
      WHEN 'IEF' THEN 10.0
      WHEN 'VNO' THEN 5.0
      WHEN 'GLD' THEN 30.0
      ELSE 0
    END / 100) * AVG(y.ror) AS weighted_return,
    (CASE sm.ticker
      WHEN 'IXN' THEN 22.0
      WHEN '000' THEN 33.0
      WHEN 'IEF' THEN 10.0
      WHEN 'VNO' THEN 5.0
      WHEN 'GLD' THEN 30.0
      ELSE 0
    END / 100) * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
        LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
      AND Ticker IN ('IXN','000','IEF','VNO','GLD')
    ) z
    ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) combined
UNION ALL

```

3721 SELECT

```

  benchmark_name AS entity,
  annualized_return,
  annualized_risk,
  sharpe_ratio
FROM uhni_1_ca.benchmark_comparison
ORDER BY sharpe_ratio DESC;

```

100% 66:3674

Result Grid Filter Rows: Search Export

entity	annualized_return	annualized_risk	sharpe_ratio
US Aggregate Bond	4.22	1.2	3.52
60/40 Portfolio	13.44	9.2	1.46
S&P 500	21.29	15.1	1.408
Optimized Portfolio - Scenario C Growth (CHOS	24.3751	21.6744	1.1246
Original Portfolio	19.6675	18.0874	1.0874

Object Info Session
No object selected

off (up to \$17) on one more. Terms apply.

Step 2E - MPT Diversification Ratio: Original 2.0199 vs Optimized 1.8372]

Administration Schemas Query 4 # A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

```

# MPT diversification check comparing original and optimized portfolio side by side

SELECT
  'Original Portfolio' AS portfolio,
  ROUND(SUM(weighted_risk), 8) AS weighted_avg_individual_risk,
  ROUND(SQRT(SUM(POWER(weighted_risk, 2))), 8) AS true_portfolio_risk,
  ROUND(SUM(weighted_risk) / SQRT(SUM(POWER(weighted_risk, 2))), 4) AS diversification_ratio
FROM (
  SELECT
    sm.portfolio_weight * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
             LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
            AND Ticker IN ('IXN','000','IEF','VNO','GLD')
    ) z
    ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) risk_table

UNION ALL

SELECT
  'Optimized Portfolio - Scenario C Growth' AS portfolio,
  ROUND(SUM(weighted_risk), 8) AS weighted_avg_individual_risk,
  ROUND(SQRT(SUM(POWER(weighted_risk, 2))), 8) AS true_portfolio_risk,
  ROUND(SUM(weighted_risk) / SQRT(SUM(POWER(weighted_risk, 2))), 4) AS diversification_ratio
FROM (
  SELECT
    CASE sm.ticker
      WHEN 'IXN' THEN 22.0
      WHEN '000' THEN 33.0
      WHEN 'IEF' THEN 10.0
      WHEN 'VNO' THEN 5.0
      WHEN 'GLD' THEN 30.0
      ELSE 0
    END * STD(y.ror) AS weighted_risk
  FROM (
    SELECT z.*, (z.p1 - z.p0) / z.p0 AS ror
    FROM (
      SELECT Ticker AS ticker, Adj_Close AS p1,
             LAG(Adj_Close, 1) OVER (PARTITION BY Ticker ORDER BY Date) AS p0
      FROM uhni_1_ca.pricing_daily_new
      WHERE Date BETWEEN '2024-06-25' AND '2026-06-25'
            AND Ticker IN ('IXN','000','IEF','VNO','GLD')
    ) z
    ) y
  JOIN uhni_1_ca.security_masterlist sm ON y.ticker = sm.ticker
  GROUP BY sm.ticker, sm.portfolio_weight
) risk_table;

```

Administration Schemas Query 4 # A1 Assignment*

Limit to 1000 rows

SCHEMAS

Filter objects

- boston_capital
- sys
- uhni_1_ca
 - Tables
 - account_dim
 - benchmark_comparison
 - customer_details
 - holdings_current
 - pricing_daily_new
 - security_masterlist
 - Views
 - Stored Procedures
 - Functions

Object Info Session

No object selected

Result Grid

Filter Rows: Search Export:

portfolio	weighted_avg_individual_risk	true_portfolio_risk	diversification_ratio
Original Portfolio	1.13939773	0.56409004	2.0199
Optimized Portfolio - Scenario C Growth	1.36536107	0.74317586	1.8372

Result Grid

Data Sources & Footnotes

Pricing Data (T5 - pricing_daily_new)

All daily OHLCV pricing data for IXN, QQQ, IEF, VNQ, and GLD was downloaded from Yahoo Finance (finance.yahoo.com) as CSV files covering June 25, 2024 to June 25, 2026 (502 trading days per ticker, 2,510 total rows). Converted from CSV to SQL INSERT statements using Python 3 with the pandas library via csv_to_sql_inserts.py and loaded into uhni_1_ca.pricing_daily_new in MySQL Workbench. Adjusted Closing Prices (Adj_Close) were used for all return calculations.

Benchmark Data (T6 - benchmark_comparison)

SPY (S&P 500): 24M CAGR from Yahoo Finance: 2024 = +24.89%, 2025 = +17.72%. CAGR = $\text{SQRT}(1.2489 \times 1.1772) - 1 = 21.26\%$. Volatility = 15.10% from PortfoliosLab.

AGG (US Aggregate Bond): 24M CAGR from Yahoo Finance: 2024 = +1.31%, 2025 = +7.19%. CAGR = $\text{SQRT}(1.0131 \times 1.0719) - 1 = 4.22\%$. Volatility = 1.20% from PortfoliosLab.

60/40 Portfolio: Blended return $(0.60 \times 21.26) + (0.40 \times 4.22)$ adjusted to 13.44%. Risk = 9.20% blended. Sharpe = 1.46 from PortfoliosLab 60/40 tracker, June 2026.

Security Master list Attributes

Expense ratios: BlackRock iShares (IEF 0.15%, IXN 0.39%), Invesco (QQQ 0.18%), Vanguard (VNQ 0.13%), State Street SPDR (GLD 0.40%). Dividend yields: IEF 3.90%, IXN 0.24%, QQQ 0.43%, VNQ 3.87%, GLD 0.00%. IEF Duration 7.70 years. All sourced from official fund issuer pages, June 2026.

Theory References

Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77-91.

Sharpe, W. F. (1966). Mutual Fund Performance. The Journal of Business, 39(1), 119-138.

Other References (Table Data)

BlackRock. (2026, June 24). iShares 7-10 Year Treasury Bond ETF (IEF) — Fund overview. BlackRock iShares. <https://www.blackrock.com/us/individual/products/239456/ishares-710-year-treasury-bond-etf>

BlackRock. (2026, March 31). iShares Global Tech ETF (IXN) — Fund overview. BlackRock iShares. <https://www.ishares.com/us/products/239750/ishares-global-tech-etf>

Invesco. (2026). Invesco QQQ ETF — Fund overview. Invesco. <https://www.invesco.com/qqq-etf/en/home.html>

MacroTrends. (2026, June 24). Invesco QQQ (QQQ) dividend yield history. MacroTrends. <https://www.macrotrends.net/stocks/charts/QQQ/invesco-qqq/dividend-yield-history>

Vanguard. (2026, June 23). Vanguard Real Estate ETF (VNQ) — Fund overview. The Vanguard Group. <https://investor.vanguard.com/investment-products/etfs/profile/vnq>

State Street Global Advisors. (2026, June 24). SPDR Gold Shares (GLD) — Fund overview. State Street SPDR. <https://www.dividend.com/etfs/gld-spdr-gold-trust/>

Note on IEF duration (7.70 years):

BlackRock. (2026, March 31). iShares 7-10 Year Treasury Bond ETF (IEF) - Fact sheet (Q1 2026).

BlackRock iShares. <https://www.blackrock.com/us/individual/literature/fact-sheet/ief-ishares-7-10-year-treasury-bond-etf-fund-fact-sheet-en-us.pdf>